Black–Scholes for Binaries: Information Clocks, Jumps and Smiles in Prediction-Market Prices

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Abstract

The fair price of a binary event contract is a bounded martingale that ends at zero or one, so its total variance is fixed by its initial price. We develop an option-pricing framework for such contracts.

In the base model, the contract is a Bachelier digital option on a latent Gaussian score. Its price solves the Black–Scholes equation with a digital payoff and has an exact transition law. The role of implied volatility is played by an *information clock*: the rate at which remaining uncertainty is resolved. We extend the model with SABR-type stochastic volatility, a general backbone and jumps, and adapt it to strike ladders, which are strips of digitals with an implied smile, and to deadline contracts, which are reduced-form hazard claims.

Calibrating on Kalshi, out of sample, we use hourly quotes on 47,500 contracts, 24,000 ladder snapshots, 5,900 deadline contracts and full order-book recordings of in-game sports markets. Prices respect the variance budget, and the clock is category-specific: uniform within games, as in Stern's Brownian model; back-loaded for weather; nearly flat before games; and rising steeply into deadlines.

Half or more of the variance arrives in jumps, even at the scale of seconds. Volatility scales with the price close to the Wright–Fisher backbone, and it is stochastic with half-lives of 7–21 hours. Ladder smiles are convex and sticky in moneyness.

Implied tail mass, implied hazards and implied variance all exceed their realized counterparts: a tail-and-time form of the favourite–longshot bias, bounded by the one-cent tick and fees.

1 Introduction

Prediction markets have become a large, regulated asset class. In the United States, an exchange such as Kalshi lists hundreds of thousands of binary contracts on sports, weather, entertainment, politics and public statements, many of them each week. Their prices are routinely read as probabilities [59, 60]. A large literature asks whether those probabilities are calibrated, documenting in particular the favourite–longshot bias [14, 32, 54, 58].

Much less is known about how these prices *move*. Option pricing begins from a model of the dynamics; the analogue for an event contract is largely missing. Such a model is needed to price derivatives of prediction-market prices, to manage the risk of quoting them, to read a probability correctly when it has just jumped, and to say what "volatility" even means for a security that must end at 0 or 1.

This paper develops such a model and calibrates it on Kalshi. The starting point is simple: the fair price of a binary contract is a bounded martingale that terminates at 0 or 1. Its total variance over its life is therefore fixed at $p_0(1-p_0)$ (Lemma 1). What a model must describe is *when*, *at which price levels* and *in what shape* this fixed variance is released.

We formalise this with a latent Gaussian score. The contract pays if the score ends above zero, so the price is a Bachelier digital option. It solves the Black–Scholes equation with a digital payoff and has an exact transition law (Proposition 1). The role of implied volatility is taken by an *information clock*: the rate at which the remaining uncertainty is resolved. We then add a SABR-type stochastic volatility with a general backbone (M1) and jumps (M2). We extend the framework to two other contract structures that Kalshi lists at scale:

- *ladders* of contracts on one numeric outcome, such as a daily high temperature or a points total. These are strips of digital options with an implied distribution, an implied volatility per strike and a smile.
- *deadline* contracts that pay if an event occurs by a date. These are reduced-form credit claims with an implied hazard term structure and a theta.

We calibrate the three branches on several data sets:

- 4.8 million hourly quote observations on 47,500 single contracts;
- 24,000 snapshots of 10,200 ladders in 9,200 events;
- 26,000 priced observations of 5,900 deadline contracts;
- full order-book recordings of 50 in-game sports markets.

Parameters are chosen on data before October 2026 and evaluated on October.

Our findings can be summarised as seven stylized facts.

1. **Prices are fair in level and in total variance.** The variance identity holds within a few percent in the broad categories.
2. **Clocks differ sharply by contract type.** In-game sports follow Stern's Brownian game almost exactly, with a uniform clock. Weather information is back-loaded into the final hours. Pre-game sports learn almost nothing. Deadline contracts carry an implied hazard that rises steeply into the deadline.
3. **Variance arrives in jumps.** Half or more of price variance comes from discrete moves, even within games at the scale of seconds. Prices drift toward resolution more slowly than the Gaussian model predicts. Outside sports the one-cent grid accounts for this; in sports, jumps do.
4. **The backbone is close to Wright–Fisher.** Price volatility scales like $[p(1-p)]^{\beta}$ with β between 0.3 and 0.8; the logit-normal model ($\beta = 1$) is rejected everywhere.
5. **Volatility is stochastic but short-lived,** with half-lives of 7–21 hours outside weather.
6. **Ladders have a smile, and it moves with the market.** The smile is sticky in moneyness rather than in strike. Implied widths and skews are close to calibrated, but implied tails are too heavy.
7. **Tails and time are overpriced.** Implied tail mass across strikes, implied hazards before deadlines and implied variance before games all exceed their realized counterparts. Prices of deadline contracts decay toward zero at only a third to two-thirds of the model rate.

These premia are the favourite–longshot bias in its cross-sectional and term-structure forms. The one-cent tick and fees bound them to a few cents per contract.

The paper proceeds as follows.

- Section 2 reviews related work.
- Section 3 describes the market and the data.
- Section 4 develops the model.
- Section 5 describes the estimators.
- Section 6 reports the results.
- Section 7 discusses their interpretation and limitations.
- Section 8 concludes.

2 Related literature

This paper draws on four literatures: prediction-market prices as probabilities, models of probability-price dynamics, option pricing and the volatility smile, and reduced-form credit models.

Prediction-market prices as probabilities. Wolfers and Zitzewitz [59] survey prediction markets and their forecasting record. Manski [45] and Wolfers and Zitzewitz [60] ask when an equilibrium price equals a mean belief. Berg et al. [10] document long-run accuracy in the Iowa Electronic Markets, and Page and Clemen [49] find that prediction markets are well calibrated except for contracts with long horizons. In betting markets the dominant departure from calibration is the favourite–longshot bias: longshots are overpriced relative to their win frequency [32, 58]. Snowberg and Wolfers [54] attribute it to misperception of small probabilities rather than risk love. Ottaviani and Sørensen [48] survey the explanations. On Kalshi, Bürgi et al. [14] use transaction data on more than 300,000 contracts. They document a favourite–longshot bias that is stronger for takers than for makers, and model it as modest belief disagreement with an overestimation of small probabilities. Rothschild and Sethi [53] study trader types and microstructure in a political prediction market.

These papers study the *level* of prices relative to outcomes. We study how prices *move*. We also show that the favourite–longshot bias reappears, in different guises, in the cross-section of strikes and in the term structure of deadlines.

Dynamics of probability prices. Stern [55] models the score difference of a game as a Brownian motion. The win probability is then a normal distribution function of the lead, scaled by the square root of time remaining: our M0 with a uniform clock. Polson and Stern [51] invert betting lines to obtain the implied volatility of a game, and Feng et al. [29] extend the approach to soccer with a Skellam process.

Archak and Ipeirotis [4] derive the volatility of a binary contract's price from latent ability diffusions. They show that volatility depends on the price and on time to expiration, and that the structural model beats GARCH in forecasting InTrade volatility. Taleb [56] uses the same construction to argue that a well-calibrated election forecast must stay close to one half when uncertainty is high.

Two recent papers are the closest to ours.

- Dalen [20] proposes a "Black–Scholes for prediction markets". It models log-odds as a jump-diffusion whose drift is fixed by the martingale condition. It describes an EM calibration that separates diffusion from jumps, and a layer of variance and correlation derivatives. Its validation is largely on synthetic paths.
- Xi et al. [61] estimate a structural volatility model on a large panel of Kalshi contracts. It combines a Wright–Fisher deadline-resolution component with a Glosten–Milgrom order-flow component [31]. They find that it forecasts volatility better than ARCH and GARCH benchmarks, that volatility peaks near even odds and rises into resolution, and that sports contracts jump while economics contracts move smoothly.

Feil and Nendel [28] derive optimal market making when the price is a transformed latent-belief diffusion; inventory-based market making goes back to Avellaneda and Stoikov [5]. Cost-function market makers [36] are an alternative price-formation mechanism, which Kalshi does not use.

Relative to this work, we make five contributions.

- We state the exact transition law of the Gaussian model and test it as a distributional restriction: the standardised innovation u must be $\mathcal{N}(0,1)$ at every horizon, given the clock.
- We estimate information clocks, backbones, stochastic volatility and jumps by contract category on out-of-sample data. At high frequency, we use simulated controls that reproduce the tick grid.
- We extend the framework to strips of contracts on a numeric outcome. There, implied distributions, smiles, smile dynamics and a variance-swap analogue are observable.
- We treat deadline contracts as a reduced-form hazard model, with implied hazard term structures, theta and event timing.
- We show that a common tail-and-time premium links these settings.

Our estimated backbone is close to the Wright–Fisher exponent of one half. This independently supports the specification of Xi et al. [61].

Option pricing, smiles and variance premia. The model is a cash-or-nothing digital option under Bachelier dynamics [6, 11, 46]. The extensions follow familiar steps:

- jumps [47];
- stochastic volatility [38];
- local and stochastic smiles [27, 34].

SABR's backbone exponent and correlation have direct counterparts in M1. Breeden and Litzenberger [13] show that a strip of option prices identifies the risk-neutral density; a strip of digitals does so directly. The distinction between sticky-strike and sticky-delta smile dynamics is due to Derman [22]; Gatheral [30] reviews smile modelling. Comparing realized with implied variance follows the variance-risk-premium literature [12, 17].

Reduced-form credit. A contract that pays if an event occurs by a deadline is the complement of a defaultable zero-coupon claim with zero recovery. Intensity models [25, 40, 43] supply the implied hazard, forward hazards and the compensated-jump martingale. The square-root intensity of Cox et al. [18] is the natural Cox specification.

Econometrics of volatility and jumps. At high frequency we use:

- volatility signature plots [2] and the noise framework of Zhang et al. [62], to separate microstructure noise from volatility;
- bipower variation [8] and threshold tests [44], to separate jumps from diffusion;
- a bid–ask bounce measure [52] and price discreteness [37], for the tick grid.

The EM decomposition of hourly moves into diffusion and jumps follows Dempster et al. [21].

3 Institutional setting and data

3.1 Kalshi event contracts

Kalshi is a US exchange for event contracts, regulated by the Commodity Futures Trading Commission as a designated contract market. Each binary contract pays \$1 if its event occurs and \$0 otherwise, and is fully collateralised. Trading takes place in a central limit order book. A bid for NO at price q is economically an offer of YES at $1 - q$, so each market has a single YES book. During our sample, prices were quoted on a one-cent grid from 1 ¢to 99 ¢.

Kalshi charges a taker fee per order of $0.07 \times C \times P(1 - P)$, rounded up, for C contracts at price P, with series-specific multipliers. Some series also carry a maker fee with coefficient 0.0175. Both fees vanish at the edges of the price range. Together with the one-cent tick, this shapes trading near 0 and 1 and matters for interpreting small departures from fair pricing (Section 7).

Contracts are organised as *series* (a recurring question), *events* (one instance, such as one game or one city-day) and *markets* (one binary leg). Several of our designs exploit this structure. A game has a winner leg and ladders of spread and total legs. A city-day of temperature has a ladder of threshold or bracket legs. A question such as "Will a model be released before T" is often listed with several deadlines.

3.2 Scope

We study *non-financial* contracts in Kalshi's seven categories:

- Sports;
- Climate and Weather;
- Entertainment;
- Mentions (whether a speaker says a phrase during an event);
- Science and Technology;
- Politics;
- Elections.

We exclude crypto, equity-index, interest-rate and economic-release contracts. Their prices are tied to traded underlyings with their own dynamics, whereas the events we study have no traded underlying, so the information structure of the contract itself is the object of interest.

3.3 Data

All data come from Kalshi's public market-data interface. We use no account-level information.

Market metadata. 1.30 million binary markets in the seven categories that closed between 15 August and 31 October 2026. For each market we have:

- its series and event;
- the rules text;
- open and close times;
- the scheduled expected expiration;
- the settlement result;
- the fee schedule.

Hourly quotes. Hourly candles with the closing best bid and ask. The base panel has 60,231 markets with the highest volume in each category, from 13 February to 31 October 2026. Two studies supplement it:

- the ladder study fetches complete ladders, 1.76 million additional hourly candles;
- the deadline study fetches the full history of deadline contracts whose deadlines fall between 1 January and 2 November 2026.

Trades. 7.3 million trade prints on 14,771 markets between 30 July and 31 October 2026, each with size, price and aggressor side.

Order-book recordings. We recorded the full WebSocket feed: order-book snapshots, deltas and trades. Receive timestamps were taken on a server in the exchange's cloud region. The recordings cover two evenings with many live games: 7 November 17:00 to 8 November 06:00 UTC, and 8 November 17:00 to 9 November 06:00 UTC. They give event-time order books for in-game game-winner markets (NBA, NCAAF, NFL, NHL).

3.4 Timing conventions and look-ahead

Three timestamps in the metadata could bias the analysis if used naively.

- *Close time.* The recorded close time of a market that resolves early reveals the event. We never use it to define a horizon.
- *Scheduled settlement.* The scheduled *expected expiration* is announced in advance, but it marks settlement, not the end of information. It lies 11–14 hours after the last weather observation and days after many speeches.
- *Deadlines.* The deadline of a "by T" contract is parsed from its rules text.

Each study therefore defines time to expiry τ from information available at the decision time:

- the single-contract study uses a per-series scheduled end of information, estimated on the estimation sample only;
- the ladder study uses leads before the scheduled settlement;
- the deadline study uses the parsed deadline.

Quotes are always taken from the last candle (or book state) strictly before the decision time.

3.5 Sample split

Every parameter, functional form and threshold is chosen on the *estimation sample*, data before 1 October 2026. Results are then reported on the *holdout*, October 2026, without re-estimation. The order-book study covers two November evenings and is purely descriptive.

4 A pricing model for binary event contracts

This section develops the pricing framework. Section 4.1 states the one property every model must respect. Sections 4.2 to 4.4 build the single-contract model in three layers: a Gaussian "Bachelier digital" with a deterministic information clock (M0), a stochastic-volatility extension with a general backbone (M1), and jumps (M2). Section 4.5 treats strips of contracts on one numeric outcome, and Section 4.6 treats contracts that pay if an event occurs before a deadline. Proofs are in Appendix A.

4.1 Setup

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\le T}, \mathbb{P})$. A binary contract pays one unit at T if an event $A \in \mathcal{F}_T$ occurs. Its fair price is the conditional probability

$$p_t = \mathbb{P}(A \mid \mathcal{F}_t), \qquad 0 \le t \le T. \tag{1}$$

We abstract from interest (collateral is posted in full and settlement horizons are short), from fees and from risk premia. Whether market prices equal p_t is an empirical question we return to in Section 7. By construction p_t is a martingale bounded in $[0,1]$ with $p_T \in \{0,1\}$. That alone fixes the total variance.

Lemma 1 (Variance identity)**.** *For any binary contract,* $\mathbb{E}[\langle p\rangle_T] = \mathbb{E}[(p_T - p_0)^2] = p_0(1-p_0)$*, where* $\langle p\rangle$ *is the quadratic variation. For prices observed at* $0 = t_0 < t_1 < \cdots < t_n = T$*,* $\mathbb{E}\sum_i (p_{t_{i+1}} - p_{t_i})^2 = p_0(1-p_0)$*.*

A model of prediction-market prices is therefore a model of *when* and *how* a fixed budget of variance $p_0(1-p_0)$ is released. It specifies the timing (a clock), the dependence on the price level (a backbone), randomness in the release rate (stochastic volatility) and its lumpiness (jumps). These are the four dimensions the empirical work measures.

4.2 M0: the Bachelier digital and the information clock

Definition 1 (M0)**.** A latent score Y satisfies $\mathrm{d}Y_t = s(t)\,\mathrm{d}W_t$ with a deterministic information flow $s(\cdot)$, and $A = \{Y_T > 0\}$. Write

$$V_t = \int_t^T s(u)^2\,\mathrm{d}u$$

for the remaining information variance, and $\lambda_t = s(t)^2/V_t$ for the *information rate*, the fraction of the remaining uncertainty resolved per unit time.

The construction goes back to Stern [55]'s Brownian model of sports scores. Its implication for the volatility of a probability price was developed by Archak and Ipeirotis [4], and Taleb [56] applies it to election forecasts.

Proposition 1 (Pricing, dynamics and transition under M0)**.** *Under M0, with Φ and φ the standard normal distribution and density:*

1. Price. $p_t = \Phi(Y_t/\sqrt{V_t})$. *As a function $p(t,y)$ it solves the backward heat equation with a digital payoff,*
$$\partial_t p + \tfrac{1}{2} s(t)^2 \, \partial_{yy} p = 0, \qquad p(T,y) = \mathbf{1}\{y > 0\}. \tag{2}$$
This is the Black–Scholes equation with zero rate and normal (Bachelier) dynamics for a cash-or-nothing payoff.

2. Dynamics. *In probit units $Z_t = \Phi^{-1}(p_t) = Y_t/\sqrt{V_t}$,*
$$\mathrm{d}Z_t = \sqrt{\lambda_t}\,\mathrm{d}W_t + \tfrac{1}{2}\lambda_t Z_t\,\mathrm{d}t, \qquad \mathrm{d}p_t = \varphi(Z_t)\sqrt{\lambda_t}\,\mathrm{d}W_t. \tag{3}$$

3. Transition. *For $h > 0$ let $R = V_t/V_{t+h} \geq 1$. Then*
$$Z_{t+h} \mid \mathcal{F}_t \sim \mathcal{N}\big(Z_t\sqrt{R},\ R-1\big), \qquad \mathbb{P}(p_{t+h} \leq x \mid p_t) = \Phi\left(\frac{\Phi^{-1}(x) - \Phi^{-1}(p_t)\sqrt{R}}{\sqrt{R-1}}\right). \tag{4}$$

Three implications organise the empirical work.

- *Clock.* The price's instantaneous volatility is $\varphi(\Phi^{-1}(p))\sqrt{\lambda_t}$: a level factor times a clock factor. With a uniform flow (s constant) the clock is $\lambda_t = 1/\tau$, with $\tau = T - t$, so volatility at a given price rises like $\tau^{-1/2}$ into expiry.
- *Test statistic.* The standardised innovation
$$u = \frac{Z_{t+h} - Z_t\sqrt{R}}{\sqrt{R-1}} \tag{5}$$
is standard normal under M0. Its variance tests the clock and its kurtosis tests the Gaussian shape.
- *Drift.* In probit units the price drifts away from zero at rate $\frac{1}{2}\lambda Z$. The martingale requires this drift to offset the convexity of Φ.

The analogues of the option Greeks follow directly. Delta is $\partial p/\partial Y = \varphi(Z)/\sqrt{V}$. The "vega" with respect to remaining variance is $\partial p/\partial V = -\frac{1}{2} Z\varphi(Z)/V$: more uncertainty pulls the price toward $\frac{1}{2}$. Theta at a fixed score is $\partial p/\partial t = \frac{1}{2} Z\varphi(Z)\lambda$: with no news, favourites drift up and longshots down. Theta is exactly offset in expectation by the diffusion term of (2).

4.3 M1: a bounded SABR model

M0 ties volatility to the level through $\varphi(\Phi^{-1}(p))$ and to time through a deterministic clock. M1 relaxes both, in the spirit of SABR [34].

Definition 2 (M1)**.**

$$\mathrm{d}p_t = \alpha_t\, c(\tau_t)\, g_\beta(p_t)\,\mathrm{d}W_t, \qquad g_\beta(p) = [p(1-p)]^\beta, \qquad \mathrm{d}\alpha_t = \nu\,\alpha_t\,\mathrm{d}B_t, \qquad \mathrm{d}\langle W, B\rangle_t = \rho\,\mathrm{sgn}(Z_t)\,\mathrm{d}t. \tag{6}$$

The parameters play the following roles.

- *Backbone β.* It nests the leading bounded diffusions:
  - $\beta = 1$ is the logit-normal model, $\mathrm{d}\operatorname{logit} p = \sigma\,\mathrm{d}W + \ldots$, the analogue of lognormal dynamics;
  - $\beta = \frac{1}{2}$ is the Wright–Fisher or Jacobi diffusion [42], the analogue of square-root dynamics;
  - the M0 backbone $\varphi(\Phi^{-1}(p))$ behaves like $p\sqrt{2\ln(1/p)}$ near zero, between the two.

  Pure Bachelier dynamics for p ($\beta = 0$) are inadmissible, because they do not keep the price in $[0, 1]$.
- *Clock $c(\cdot)$.* The deterministic information schedule.
- *Vol of vol ν.* How strongly the information rate itself fluctuates.
- *Correlation ρ.* The factor $\operatorname{sgn}(Z)$ makes the specification invariant to relabelling YES and NO. With $\rho < 0$, volatility rises when the price moves toward $\frac{1}{2}$; with $\rho > 0$, it rises as the price moves toward resolution.

Lemma 1 holds for every specification of M1. Stochastic volatility redistributes the variance budget across time and paths; it does not enlarge it.

4.4 M2: jumps

Information often arrives discretely: scores, rulings, releases, the start of a speech. M2 adds compensated jumps to the latent score,

$$\mathrm{d}Y_t = s(t)\,\mathrm{d}W_t + J_t\,\mathrm{d}N_t, \tag{7}$$

where N is a counting process with intensity $\kappa(t)$ and J an independent jump size. With Gaussian jumps $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$, a constant intensity and n jumps remaining with Poisson probability π_n, the price is a Merton [47]-type mixture,

$$p_t = \sum_{n \geq 0} \pi_n \, \Phi\left(\frac{Y_t + n\mu_J}{\sqrt{V_t + n\sigma_J^2}} \right). \tag{8}$$

Between jumps, the drift that keeps p a martingale is the compensator of the expected jump, not the diffusive convexity term. The M0 link between local variance and outward probit drift (Proposition 1, item 2) therefore breaks when a share of the variance arrives in jumps. We use this in Section 6.2.

4.5 Strips of digitals on one outcome: implied distributions and smiles

Many contracts are listed as ladders on one numeric outcome X_T, such as a temperature, a points total or a score. Each leg pays $\mathbf{1}\{X_T > K\}$ (or $\mathbf{1}\{K_1 < X_T \leq K_2\}$ for brackets). The survival function $S_t(K) = \mathbb{P}(X_T > K \mid \mathcal{F}_t)$ is then the price of a strip of digital options, and it identifies the conditional distribution of X_T. This is the digital counterpart of Breeden and Litzenberger [13].

Definition 3 (Implied median, implied volatility, moneyness)**.** The implied median is $F_t = S_t^{-1}(\frac{1}{2})$. The Bachelier-digital implied volatility of strike K solves

$$S_t(K) = \Phi\big((F_t - K)/\sigma_K\big),$$

and moneyness is $m = (K - F_t)/\sigma_{\mathrm{ATM}}$. The smile is the map $m \mapsto \sigma_K/\sigma_{\mathrm{ATM}}$, summarised by

$$\sigma_K/\sigma_{\mathrm{ATM}} - 1 = a + s\,m + c\,m^2,$$

with skew s and curvature c.

Proposition 2 (Shape of the digital smile)**.**

1. *If $X_T \mid \mathcal{F}_t$ is Gaussian, the smile is flat: $\sigma_K \equiv \sigma_{\mathrm{ATM}}$.*
2. *If $X_T \mid \mathcal{F}_t$ is symmetric with tails heavier than the Gaussian of the same interquartile range, then $\sigma_K > \sigma_{\mathrm{ATM}}$ for every strike beyond the quartiles: the smile turns up in the wings. For Student-t outcomes, $\sigma_K/\sigma_{\mathrm{ATM}}$ increases in $|m|$ (verified numerically; Appendix A).*
3. *Skewness tilts the smile: right skew makes σ_K larger above F than below it.*

Two further concepts carry over from equity options. The first is smile dynamics: a smile is *sticky-strike* if σ_K at a fixed K does not respond to moves in F, and *sticky-moneyness* if σ_K depends on K only through m [22]. The second is a variance-swap analogue. If X_T is Gaussian given $\mathcal{F}_t$, each leg is an M0 contract with $Y = X - K$ and $V_t = \sigma_{\mathrm{ATM}}(t)^2$. The ladder's at-the-money term structure is then an *implied information clock*, and the implied median is a martingale with

$$\mathbb{E}\big[(F_{t+h} - F_t)^2\big] = \mathbb{E}\big[\sigma_{\mathrm{ATM}}(t)^2 - \sigma_{\mathrm{ATM}}(t+h)^2\big]. \tag{9}$$

Comparing the two sides is the prediction-market analogue of comparing realized and implied variance [12, 17].

4.6 Deadline contracts: a reduced-form hazard model

A large class of contracts pays if an event occurs *by* a date T: a product release, a statement, an elimination, a weather system forming. We model the event time ϑ with an intensity h_t, as in reduced-form credit models [25, 40, 43].

Proposition 3 (Hazard pricing)**.** *On $\{\vartheta > t\}$ the price is*

$$p_t = 1 - \mathbb{E}_t\Big[\exp\Big(-\int_t^T h_u\,\mathrm{d}u\Big)\Big].$$

1. Deterministic hazard. $p_t = 1 - e^{-H(t,T)}$ *with $H(t,T) = \int_t^T h_u\,\mathrm{d}u$. Before the event the price decays deterministically,*
$$\frac{\mathrm{d}p_t}{\mathrm{d}t} = -h_t\,(1 - p_t), \tag{10}$$
and at ϑ it jumps to 1. The decay is exactly the compensator of the jump, so p is a martingale.
2. Implied average hazard. $\bar{h}_t = -\ln(1 - p_t)/(T - t)$.
3. Term structure. *For a strip with deadlines $T_1 < T_2$ written on the same event, the forward hazard is*
$$h(T_1, T_2) = -\frac{\ln\big[(1 - p_t(T_2))/(1 - p_t(T_1))\big]}{T_2 - T_1} \geq 0,$$
so $p_t(T_1) \leq p_t(T_2)$ is a static no-arbitrage condition.

Table 1: Model objects and their option-pricing analogues.

Prediction-market object	Symbol	Analogue
Remaining information variance	V_t	total implied variance $\sigma^2\tau$
Information rate (clock)	$\lambda_t = s_t^2/V_t$	forward variance per unit of total variance
Probit price	$Z = \Phi^{-1}(p)$	normalised moneyness d_2
Backbone	β in $[p(1-p)]^\beta$	CEV exponent / SABR β
Vol state, vol of vol, correlation	α, ν, ρ	SABR α, ν, ρ
Ladder implied median	F	forward
Ladder implied volatility	σ_K, σ_{ATM}	Bachelier implied volatility
Deadline implied hazard	$\bar{h}$	credit spread / default intensity
Decay with no event	$-h(1-p)$	carry of a short credit position

A Cox specification, in which h is itself a diffusion (for example the square-root intensity of 18), adds news-driven diffusion to the deterministic decay and the resolution jump. We call the deterministic decay in (10) the contract's theta.

Summary. The three contract types share one structure: a latent state, an information clock that releases a fixed variance budget, and jumps. They differ in the latent state, which is a Gaussian score, a numeric outcome, or an event time. Table 1 maps the objects to their option-pricing analogues.

5 Empirical methodology

The three studies share a set of conventions, collected here.

Price proxy. The price is the mid-quote. Each study admits it only when the quoted spread is narrow: at most 3–5 ¢, depending on the study. Away from the edges the mid is then within a cent or two of any reasonable estimate of the fair price. Near 0 and 1, the one-cent grid and the price floor make probit and logit transforms of the mid unstable. Studies therefore either restrict the price range or model the grid explicitly.

Inference. Observations within an event (one game, one city-day, one question) share information shocks. All standard errors are cluster-robust by event [15], and t-statistics are reported with event clustering unless stated otherwise. Each study reports the number of clusters.

Distributional tests. Distributional statements are tested with probability integral transforms (PIT) of realized outcomes under the model or implied distribution [24]. Interval-censored outcomes, where only the settled bracket is known, are handled by splitting the PIT over the interval. Volatility forecasts are compared with the QLIKE loss, which is robust to noise in the realized-variance proxy [50]. MSE is reported alongside it.

Microstructure controls. Ticks, spreads and price clipping mechanically affect higher-moment and drift statistics. Where they could, the studies simulate the null model on the same one-cent grid with the observed spreads and the same filters, and report every statistic for both the data and this matched control. A finding is attributed to the market only where it differs from the control.

Economic significance. Some sections translate a mispricing into the return of a simple rule. Those rules use executable quotes (the bid or the ask, not the mid) and Kalshi's fee schedule. Positions are held to settlement. Thresholds are chosen on the estimation sample. These calculations measure the size of departures from fair pricing; they are not proposed as trading strategies.

Multiple comparisons. Results are reported across categories, horizons and leads. We emphasise patterns that are consistent in sign across the estimation and holdout samples, and across related cells, over individual significant cells.

5.1 Single-contract dynamics

5.1.1 Sample, clock time and look-ahead guards

Hourly panel. We use the hourly candles of Section 3 for the settled markets of the seven non-financial categories, February to October 2026. Kalshi omits a candle when nothing changed, so each market is placed on a regular hourly grid from its first candle to its last candle at or before `close_time`, with the closing yes bid and ask forward-filled. Candles ending after the close are dropped. This gives 4,828,986 market-hours on 47,505 markets. Sports rows are split into *pre-game* (the step ends before the scheduled start, parsed from the ticker or set to expected expiration minus 3 h) and *in-game*.

A step is $(t, t+h]$ with $h = 1$ h. The decision quote must have a spread of at most 4 ¢ and a mid in $[0.03, 0.97]$; the end quote must be two-sided with a spread of at most 10 ¢. This measurement filter removes 0.4–5.2% of otherwise valid steps. Mids are clipped to $[0.005, 0.995]$ before taking $Z = \Phi^{-1}(p)$.

Time to the scheduled end. The model needs τ, the time until information stops arriving (Section 4). `close_time` cannot be used: for markets that close early it is look-ahead, since an early close usually reveals the outcome. `expected_expiration_time` is the settlement schedule and can lag the information by days. We therefore set $T =$ `expected_expiration_time` $- o_s$ per series. The offset o_s is the modal (expiration − close) among the series' markets that closed before 1 October 2026 if at least half share it, and otherwise their median over NO-settled markets, which run to the scheduled end. Series without such markets have no schedule and are excluded; this removes all October Elections series, so Elections has no holdout. Sports use $T =$ `expected_expiration_time`, about the start plus 3 h.

On NO-settled markets the median of `close_time` $- T$ is 0 in every non-sports category in the estimation sample, and in the holdout everywhere except Mentions (+10.4 h). Steps ending less than 30 minutes before T are dropped. The estimation sample (before 1 October 2026) contains steps that end before 1 October and the holdout (October 2026) contains steps that start on or after it. All parameters and tuning choices use the estimation sample only, and t-statistics are clustered by event (Section 5).

5.1.2 Clock, innovations and the variance identity

Let $L_t = \int_t^{t+h} \lambda_s \, \mathrm{d}s = \ln(V_t / V_{t+h})$ be a step's clock increment, so that M0 gives $Z_{t+h} \mid Z_t \sim \mathcal{N}(Z_t e^{L_t/2},\, e^{L_t} - 1)$ (Section 4). Treating the observed mid as the fair price plus noise of variance $k s_t^2$ (s_t the spread), we maximize

$$\ell = \sum \log \varphi_{\mathcal{N}}\Big(\tilde{Z}_{t+h};\; \tilde{Z}_t e^{L_t/2},\; e^{L_t} - 1 + k\, s_t^2 / \varphi(\tilde{Z}_t)^2\Big) \tag{11}$$

over the clock and k. Tildes denote observed mids.

The candidate clocks are uniform ($\lambda = 1/\tau$, $L = \ln[\tau/(\tau - h)]$), scaled (a/τ), power ($a\tau^{-b}$, $L = a[\tau^{1-b} - (\tau - h)^{1-b}]/(1-b)$), piecewise constant over τ buckets with edges $2, 4, \ldots, 1024$ h, and the power and piecewise forms times 24 hour-of-day factors with mean one ("×hour"). Each is fitted per category by L-BFGS on up to 200,000 estimation steps. The BIC-best form is used downstream, and forms are compared by holdout mean negative log-likelihood per step relative to uniform.

The standardized innovation

$$u_t = \frac{\tilde{Z}_{t+h} - \tilde{Z}_t e^{L_t/2}}{\sqrt{e^{L_t} - 1 + k\, s_t^2/\varphi(\tilde{Z}_t)^2}} \tag{12}$$

is $\mathcal{N}(0,1)$ under M0. We report its variance, excess kurtosis and tail frequencies.

The *probit-drift* test regresses $\tilde{Z}_{t+h} - \tilde{Z}_t$ on $\tilde{Z}_t(e^{L_t/2} - 1)$ for steps with $L_t \le 2$. M0 predicts a slope of one, the outward drift that keeps p a martingale. An IV version instruments with the previous hour's $\tilde{Z}$ to remove errors-in-variables bias from quote noise.

The *variance identity* is checked per market: $\sum(\Delta p)^2$ over two-sided quotes (spread at most 4 ¢) sampled every $k \in \{1, 6, 24\}$ h, plus the final move $(y - p_{\text{last}})^2$, compared with $p_0(1 - p_0)$.

5.1.3 Backbone, stochastic volatility, leverage and jumps

Backbone. For each backbone g we fit by QLIKE

$$\mathbb{E}[(\Delta p)^2 \mid p, \tau] = c_{j(\tau)}^2\, g(p)^2/g(1/2)^2 + k\, s^2, \tag{13}$$

with one scale per τ bucket j. The candidates are the M1 power family $g = [p(1-p)]^\beta$, logit-normal ($\beta = 1$), Jacobi ($\beta = 1/2$) and the probit backbone $\varphi(\Phi^{-1}(p))$ of M0; they are compared by holdout QLIKE relative to probit. A model-light check regresses log cell means of $(\Delta p)^2$, Roll-corrected as $\mathbb{E}(\Delta p_t)^2 + 2\mathbb{E}(\Delta p_t \Delta p_{t+1})$ [52], on τ-bucket fixed effects and $2\beta \log[p(1-p)]$, with an event-level Poisson bootstrap (100 draws).

Volatility clustering. We estimate the within-market autocorrelation of $|u_t|$ at lags 1–168 h. It is computed pooled and, removing static differences between markets, demeaned by market (markets with at least 48 steps). We fit $\text{Corr}(|u_t|, |u_{t+k}|) = A\phi^k$ over $k \le 24$ and report the half-life $\ln 0.5/\ln\phi$. Under log-normal stochastic volatility [57], $A = (x-1)/(\frac{\pi}{2}x - 1)$ with $x = e^{\sigma_h^2/4}$ and $\sigma_h^2 = \text{Var}(\log \alpha^2)$. This gives the M1 vol-of-vol $\nu = \frac{1}{2}\sqrt{\sigma_h^2(1-\phi^2)}$ per $\sqrt{\text{hour}}$, which we report per $\sqrt{\text{day}}$.

Leverage. The sign of ρ, in the $\text{sgn}\, Z$ convention of M1, comes from

$$\text{Corr}\left(\text{sgn}(Z_t)\, u_t,\ \tfrac{1}{6}\textstyle\sum_{j=1}^{6} |u_{t+j}|\right), \tag{14}$$

on steps with $p \in [0.1, 0.9]$ and u winsorized at ± 8. A negative value means volatility rises after moves toward 1/2. A price-unit version replaces u by Δp over the fitted backbone volatility.

Jumps. Following M2, an EM [21] fits the zero-mean scale mixture

$$u \sim (1-\pi)\,\mathcal{N}(0, s_d^2) + \pi\,\mathcal{N}(0, s_d^2 + s_J^2) \tag{15}$$

to steps in which the mid moved; hours without a change are reported as the zero-move share. The jump share of variance is $\sum \gamma_i(\Delta p_i)^2 / \sum(\Delta p_i)^2$, with γ_i the posterior jump probability. The mixture is a random clock burst, which keeps p an exact martingale.

5.1.4 Variance forecasts

The target is $RV_{t,H} = \sum_{j<H}(\Delta p_{t+j})^2$ for $H \in \{1, 6\}$ h, on decision rows with at least 24 h of history. We compare five forecasts:

- HIST: the trailing mean of $(\Delta p)^2$ over $W \in \{6, 24, 72\}$ h, times H;
- M0: the closed-form transition variance under the fitted clock;
- BB: the backbone local variance of Equation (13);
- M1: the M0 variance times $\hat{\alpha}_t^2 = (1 - w) + w\,\mathrm{EWMA}_{\mathrm{hl}}[(\Delta p)^2/v_1]$ over the market's past steps, with v_1 the one-step M0 variance plus noise, hl $\in \{3, 12, 48\}$ h and $w \in \{0.25, \ldots, 1\}$;
- M1BB: BB times $\hat{\alpha}_t^2$.

Each forecast is calibrated on the estimation sample as $\kappa \cdot \mathrm{base} + c_0 s^2 H + c_1$ by QLIKE, with W, hl and w chosen there. Holdout accuracy is measured by QLIKE, $\log f + RV/f$ [50], and by MSE, with event-clustered t-statistics on per-row loss differences [23].

5.1.5 High-frequency in-game sample

For the nights of 7–8 and 8–9 November 2026 (17:00 to 06:00 UTC) we replay every recorded order-book snapshot and delta of the public Kalshi websocket for game-winner series. Each market's book is reduced to its top, persists across hourly files and is reset by snapshots. We keep one leg per game, the first alphabetically excluding ties. Its in-game window runs from expected expiration minus the league's game length to expected expiration plus 1 h. The window is cut at the first decided quote (mid at most 2 ¢ or at least 98 ¢, spread at most 10 ¢) and 10 minutes before T. Legs with at least 30 minutes remaining are kept: 50 legs (NBA 14, NCAAF 14, NFL 10, NHL 12), 6,965 in-game minutes and 267,422 top-of-book changes.

The mid is sampled as of each point of an h-second grid, $h \in \{1, 5, 15, 60, 300\}$. Steps do not overlap and use the uniform clock with τ in seconds. The start spread is at most 3 ¢ (2 ¢ as a robustness check), the start mid is in $[0.03, 0.97]$ and the end spread is at most 10 ¢; t-statistics cluster by game.

Signature and jumps. The variance signature is $RV(h)/T$ per in-game hour for h from 1 to 600 s on the same windows [2]. Under i.i.d. noise of standard deviation η in the mid,

$$\mathbb{E}[RV(h)]/T = \sigma^2 + 2\eta^2/h, \tag{16}$$

which we fit across h [1, 62].

At 5, 10 and 15 s we compare realized variance with bipower variation, $BV = \mu_1^{-2} \sum_i |r_i||r_{i-1}|$ with $\mu_1 = \sqrt{2/\pi}$, and report the jump share $(RV - BV)/RV$ [8, 9, 39]. Threshold jumps, following Lee and Mykland [44], have $|u_i| > 4\hat{\sigma}_i$ and $|\Delta p_i| \geq 2$ ¢, where $\hat{\sigma}_i$ is the bipower scale of u over the preceding 30 minutes, floored at 0.25.

For a fixed score increment, Stern's Brownian game model predicts a jump size proportional to $\varphi(Z)/\sqrt{\tau}$ [51, 55]. We regress $\log|\Delta p|$ on that quantity over the detected 10-second jumps, with league fixed effects; the predicted slope is one. Leverage at 1, 2 and 5 minutes uses Equation (14) with the next five steps.

5.1.6 Tick-grid control simulations

Matched high-frequency control. For each leg we simulate an M0 path per second under the uniform clock from the leg's first mid. We quote it on the 1 ¢ grid with the leg's observed spread, bid $= \text{round}(100p - s/2)$ ¢ and ask $=$ bid $+ s$, apply the same cuts, and compute the same statistics. The control therefore has M0, ticks and the real spreads, but no jumps and no stochastic volatility.

Hourly drift control. M0 paths are simulated on each market's real hourly grid with the category's fitted clock, in three versions:

- *true*: not quoted;
- *tick*: quoted on the grid with the real spreads;
- *stale*: quoted on the grid but re-quoted only in hours with a real candle.

To locate any shortfall we refit the drift regression inside $|u| \le 2$, a window centred on M0's mean that biases the slope toward one, and inside $|\Delta Z| \le 2\,\text{sd}$, centred on zero, which biases it toward zero. On $|Z| \ge 0.5$ we also compute the local second-order martingale ratio $\mathbb{E}[\text{sgn}(Z)\Delta Z]/\mathbb{E}[\frac{1}{2}|Z|(\Delta Z)^2]$, which is one when p is a martingale, whatever the clock.

5.2 Strike ladders

A ladder is the set of markets of one Kalshi event that pay on the same numeric outcome X: the daily maximum or minimum temperature at a station, the total points or the margin of a game, an approval average, a count of posts. Under the strip-of-digitals reading of Section 4, each leg prices one point (or one interval) of the survival function $S_t(K) = \mathbb{P}_t(X_T > K)$, and the ladder as a whole implies a distribution for X_T. This subsection describes how we recover that distribution from quotes, how we summarise its shape and its dynamics, and how we test it against the realised outcome. Inference follows the conventions of Section 5: parameters and rules are chosen on the estimation sample (events expiring before 1 October 2026), the holdout is October 2026, and standard errors are clustered by event.

Ladders and legs. We take every settled event whose markets all carry a numeric strike in their published subtitle, in five groups: daily high and low temperature (six-leg bracket sets), sports totals (full game and periods), sports team totals, sports point spreads, and numeric ladders in politics, entertainment, science and technology and other climate series. Player props, exact-score and mention markets are excluded. A leg pays YES iff $X_T \in (\ell, h]$, and the subtitle determines the edges. In bracket events (any "between" or exact-value leg) the underlying is discrete and edges move half a grid step d outwards, with d half the smallest gap between adjacent brackets:

$$\begin{gathered}\text{“}a \text{ to } b\text{”} \mapsto (a-d,\, b+d], \qquad \text{“}{\ge}a\text{”} \mapsto (a-d, \infty), \qquad \text{“}{>}a\text{”} \mapsto (a+d, \infty), \\ \text{“}{<}a\text{”} \mapsto (-\infty,\, a-d], \qquad \text{“}{\le}a\text{”} \mapsto (-\infty,\, a+d].\end{gathered} \tag{17}$$

Threshold-only ladders use half the strike resolution, which is zero for half-point sports lines. A spread ladder puts both teams on one margin $X = A - B$ ("B wins by over k" is $X < -k$); team totals form one ladder per team. The realised outcome is known only as the interval $X_T \in (L, U]$ implied by which legs settled YES. Ladders whose settlements are inconsistent with any single value of X_T, and events whose expected expiration lies more than three days after their close (mostly elections), are dropped. Before quote filters this gives 33,002 ladders with 326,286 legs.

Snapshots and filters. For non-weather ladders we take snapshots 48, 24, 6 and 1 hours before the scheduled expected expiration. For sports this time is close to the scheduled end of the game, so the 48–6 hour snapshots are pre-game and the 1 hour snapshot is in play. Weather markets close at the end of the measurement day D in local standard time, and we sample at 12:00 on $D-1$ and at 00:00, 08:00, 12:00 and 16:00 on D, that is 36, 24, 16, 12 and 8 hours before the close. The quote of each leg is the bid and ask close of the last hourly candle that ended at or before the snapshot (at most two hours old), for legs that are open at the snapshot. A leg enters the fit only if its spread is at most 5 ¢; a bid of zero is allowed for a wing whose ask is at most 5 ¢.

Survival points. A "greater" leg with mid q gives $S(\ell) = q$ and a "less" leg gives $S(h) = 1 - q$, each with weight $1/\max(\text{spread}, 1\,¢)^2$; legs at the same strike are averaged. A bracket set enters only if every leg passes the filter and the edges are contiguous; then S at each interior edge is the summed mid of the legs above it divided by the summed mid of all legs, which removes the overround multiplicatively. Weighted isotonic regression [7] makes S non-increasing in K. A snapshot is kept if it has at least three survival points and the implied median lies inside the quoted strikes.

Implied median, width and moneyness. Let $z_i = \Phi^{-1}(S_i)$ with S clipped to $[0.005, 0.995]$. The implied median F is the strike at which z crosses zero, by linear interpolation between the two quoted strikes that straddle it. For the ATM width we use the implied interquartile range,

$$\sigma_{\text{ATM}} = \frac{K_{0.25} - K_{0.75}}{2\,\Phi^{-1}(0.75)}, \qquad S(K_q) = q, \tag{18}$$

with K_q interpolated linearly in z. We fall back to the local slope between the two strikes straddling F when the quartiles are not both inside the quoted range. Under the Bachelier digital of Section 4 the two definitions coincide. In data they do not: adjacent strikes on a football ladder can be one point apart, and point masses at key margins dominate the local slope, whereas the interquartile range averages over them. The interquartile definition is used in 84% of snapshots overall and in 76–98% of snapshots in each large group. Per strike we compute the Bachelier implied volatility and the moneyness

$$\sigma_K = \frac{F - K}{\Phi^{-1}\big(S(K)\big)}, \qquad m = \frac{K - F}{\sigma_{\text{ATM}}}, \qquad v_K = \frac{\sigma_K}{\sigma_{\text{ATM}}}, \tag{19}$$

keeping strikes with $|z| \geq 0.25$ whose S is strictly inside the clip. Spread ladders are reflected ($X \mapsto -X$) when $F < 0$, so that X is the favourite's margin; the decision uses prices only.

Smile regression. Within each group and lead we estimate by pooled least squares

$$v_K - 1 = a + s\,m + c\,m^2 + e_K, \qquad |m| \leq 3.5, \tag{20}$$

excluding the two strikes that straddle F, whose volatilities equal the local slope by construction. Here s is the skew ($s > 0$: upper-wing volatilities exceed lower-wing ones) and c the curvature ($c > 0$: implied tails fatter than a normal with the ATM width). We also fit (20) with $a = 0$ snapshot by snapshot when at least three off-ATM strikes are available.

Density shape. To compare the implied and realised shapes in common units, we fit the skew-t distribution of Jones and Faddy [41], $G(x; a, b) = I_{y(x)}(a, b)$ with $y(x) = \frac{1}{2}\big(1 + x/\sqrt{a + b + x^2}\big)$ and I the regularised incomplete beta function. Its left and right tails decay like Student-t tails with $2a$

and $2b$ degrees of freedom. The implied shape is fitted to the pooled standardised survival points, $S(m) = 1 - G((m-\mu)/\varsigma; a, b)$, by least squares on the logit scale with S clipped to $[0.01, 0.99]$. The realised shape is the interval-censored maximum likelihood fit of $z_T = (X_T - F)/\sigma_{\text{ATM}}$. We report two unit-free summaries of each fit: quantile (Bowley) skewness $(q_{0.9} + q_{0.1} - 2q_{0.5})/(q_{0.9} - q_{0.1})$ [33] and tail weight $(q_{0.975} - q_{0.025})/[2.906\,(q_{0.75} - q_{0.25})]$, which equals one for a normal. The implied summaries carry 90% intervals from 30 cluster-bootstrap replications over events.

Smile dynamics. Take consecutive snapshots t and u (the next shorter lead) of the same ladder in its original orientation, and write σ_t and σ_u for σ_{ATM} at the two snapshots. Define $\Delta F_n = (F_u - F_t)/\sigma_t$ and $\Delta\sigma_n = (\sigma_u - \sigma_t)/\sigma_t$. In the terminology of Derman [22], a *sticky-strike* smile is fixed in K, so the ATM point slides along it and $\Delta\sigma_{\text{ATM}} \approx s\,\Delta F$, since $\partial\sigma_K/\partial K = s$ at the money. A *sticky-moneyness* smile is fixed in m, so $\Delta\sigma_{\text{ATM}} = 0$ apart from level changes, while the volatility at a fixed strike moves by $-\sigma'(m_K)\Delta F$, with $\sigma'(m) = s + 2cm$. We estimate, with fixed effects for each group and lead pair,

$$\Delta\sigma_n = b\,\Delta F_n + e, \tag{21}$$

$$\frac{\sigma_{K,u} - \sigma_{K,t}}{\sigma_t} = \gamma\left[-(s + 2c\,m_{K,t})\,\Delta F_n\right] + \phi\,v_{K,t}\,\Delta\sigma_n + e. \tag{22}$$

Sticky strike predicts $b = s$ and $\gamma = 0$; sticky moneyness predicts $b = 0$ and $\gamma = 1$. In (22) (s, c) are either the pooled smile of the group and lead, which has no measurement error at the snapshot level, or the snapshot's own fit, which is noisy and biases γ towards zero. The ATM regression is also run with $s\,\Delta F_n$ as a regressor, whose coefficient δ is 1 under sticky strike. For the backbone (Hagan et al. [34]; Section 4) we regress $\Delta\log\sigma_{\text{ATM}}$ on $\Delta\log F$ within ladders, for groups whose underlying is positive, and also run the cross-sectional analogue, $\log\sigma_{\text{ATM}}$ on $\log F$ with series-by-lead fixed effects. An exponent β of 0 corresponds to a normal backbone, $\frac{1}{2}$ to a square-root (Poisson-count) backbone and 1 to a lognormal one.

ATM term structure. Under a uniform information clock the remaining variance is proportional to the time left, so $\sigma_{\text{ATM}} \propto \tau^{1/2}$ (Section 4). We estimate the exponent η in $\log\sigma_{\ell j} = \alpha_\ell + \eta\log\tau_j + e$, where $\sigma_{\ell j}$ is σ_{ATM} of ladder ℓ at lead τ_j and α_ℓ are ladder fixed effects. Sports use pre-game leads only.

Calibration against the realised outcome. *PIT.* Because X_T is interval-censored, its probability integral transform is the interval $[1 - S(a'),\ 1 - S(b')]$, where $a' \le L$ and $b' \ge U$ are the nearest quoted strikes (0 and 1 beyond the quoted range). Spreading each interval uniformly gives a fractional histogram that is uniform under calibration [19, 24]. We report the share of mass in the outer two deciles (0.2 under calibration) with t clustered by event. *Width ratio.* We fit $(X_T - F)/\sigma_{\text{ATM}} \sim \mathcal{N}(\mu, \lambda^2)$ by interval-censored maximum likelihood with a cluster sandwich; $\lambda < 1$ means the implied ATM width is too wide. *Tail mass.* For strikes with $m \ge k$ (upper wing) we compare the implied $S(K)$ with the realised $\mathbf{1}\{X_T > K\}$, and for $m \le -k$ the implied $1 - S(K)$ with $\mathbf{1}\{X_T \le K\}$. The statistic z_{H_0} treats the implied probabilities as true: event means of realised minus implied, variance equal to the event mean of $S(1-S)$, events independent. Unlike a clustered t it remains defined when no tail event occurs.

Implied versus realised variance between snapshots. If F_t were the conditional mean of X_T and σ^2_{ATM} its conditional variance, the law of total variance would give $\mathbb{E}_t[(F_u - F_t)^2] = \mathbb{E}_t[\sigma_t^2 - \sigma_u^2]$.

This motivates the ratio

$$\mathrm{RV/IV} = \frac{\sum(\Delta F_n)^2}{\sum\left[1-(\sigma_u/\sigma_t)^2\right]}, \tag{23}$$

summed over pairs of a group and window, with a 90% cluster-bootstrap interval (200 replications over events). It is a variance-swap analogue of the variance risk premium [12, 17]: a ratio below one means the implied distribution narrows faster than its median moves. With a median and an interquartile width the identity is exact only for symmetric, normal shapes, and we read the ratio descriptively.

Economic significance. We record the outcome of selling a leg's tail side (buying the other side at the snapshot ask, Kalshi taker fee by series, one contract, held to settlement), gated either by model moneyness or by price alone. Because these wins come from rare tail events, we report a t against settlement risk at the market price, t_{risk} (per-contract standard deviation $\sqrt{p(1-p)}$, summed within events), in place of the clustered t.

5.3 Deadline contracts

A deadline contract pays one dollar if a specified occurrence happens after issuance and before a stated date T; the exchange may close it as soon as the occurrence is confirmed. Under the model of Section 4 its price is one minus a survival probability. We measure three objects: the implied average hazard $\bar{h}_t$, its term structure across deadlines, and the no-event decay (theta) of the price.

Sample and classification. Deadline contracts live for weeks or months, so we extend the settled-market table of Section 3 backwards series by series through the public historical endpoint. The extension covers 237 series with a deadline contract in the table and 1,450 further non-financial one-off, custom or periodic series whose title contains a deadline-type word ("before", "out", "release", "visit", "eliminated" and similar). Series are selected on titles, never on outcomes. Contracts are classified from their rules text alone. A candidate's rules contain a deadline phrase ("before D", "by D", "through D", "between D_1 and D_2", optionally with a clock time), and candidates fall into four hazard types (Table 2):

- *events*: single occurrences, such as a release, resignation, visit or launch;
- *mentions*: weekly or monthly windows in which a named speaker must say a word;
- *counts*: first-passage thresholds ("at least N ... before D");
- *eliminations*: reality-show eliminations, resolved at scheduled episodes.

We exclude competing legs, which resolve NO early when another candidate succeeds; quantities measured at T; scheduled votes; and scheduled shows and elections. A contract is in scope when its parsed deadline lies in (1 January, 2 November 2026], it settled YES or NO, and it was open after 1 January 2026. The upper bound is the key look-ahead guard: a contract whose deadline lies beyond the data end would appear only if it had already resolved YES. The sample has 5,872 contracts, 5,761 of them with hourly quotes.

Table 2: Classification of deadline contracts

Type	Cand.	Series	In scope	Clusters	YES rate	Example
Event	3,505	700	2,180	548	0.306	Google releases next Gemini Pro before 16 Oct
Mention	2,567	8	1,940	71	0.588	Trump says "Epstein" before 2 Nov
Elimination	1,485	8	1,449	88	0.109	Contestant eliminated from DWTS before 28 Oct
Count	504	31	303	52	0.399	Trump endorses ≥ 5 people this week
Excluded						
Competing legs	2,401	64				Player's next team before 1 Dec: Washington
Measured at T	1,014	25				Weekly token total above a level
Scheduled vote	250	10				Senator votes for a cloture motion
Scheduled show or election	36	7				Artist performs a song at a named show

Candidates (Cand.) are non-financial single-leg contracts whose rules contain a deadline phrase. In scope: parsed deadline in (1 January, 2 November 2026], settled YES or NO, open after 1 January 2026. Clusters: distinct exchange event tickers among in-scope contracts (the clustering unit). YES rate: share of in-scope contracts that resolved YES. Examples paraphrase contract titles.

Deadlines, quotes and look-ahead guards. The deadline is parsed from the rules: "before D" ends at 00:00 US Eastern on D; "by", "through" and "between ... and D" include D; an explicit clock time overrides. The window start s is an explicit "after D"/"from D" date or, failing that, issuance. Close and settlement times are never used, since they reveal an early YES. Of contracts resolving NO, 97.6% closed within 26 hours of the parsed T (median offset -0.02 hours). Contracts resolving YES closed more than six hours early in 90.1% of events and 81.8% of mentions, but only 31.6% of eliminations. Decisions are taken at $t = T - L$ for leads L of 1, 3, 6, 12, 24, 48, 72, 120, 168, 240, 336, 504, 720, 1080 and 1440 hours. The quote is the bid and ask of the last hourly candle that ended strictly before t, at most two hours old. An observation is *priced* when the spread is at most 10 ¢ (fixed in advance). The estimation sample (before 1 October 2026) holds contracts with T before 1 October 2026. The holdout (October 2026) holds decisions taken in October with $T \leq 2$ November. Straddling contracts enter pooled tables only. The 26,290 priced observations cover 5,070 contracts in 709 event clusters: 22,736 in the estimation sample and 2,076 in the holdout. Measures that condition on "no event yet" use an upper bound on the event time, τ_E: the earlier of the YES close and the end of the first hourly candle of the final run of mids of at least 97 ¢. τ_E never enters calibration or trading statistics. Conditioned measures are restricted to mids below 0.9, so that a confidently priced contract is not flagged as resolved before its event.

Implied hazards and strips. $\bar{h}_t = -\ln(1 - p_t)/(T - \max(t, s))$, with p_t the mid. A strip is a set of event or mention contracts with identical rules apart from dates. Legs are compared only while both are open and their windows have started, and only if the later deadline does not start later than the earlier one. Where a window starts at issuance, this assumes that a later deadline issued no later than an earlier one covers its window; without such nesting, monotonicity in T is not an arbitrage bound. From one daily snapshot per strip (16:00 UTC, spread at most 10 ¢) we bootstrap forward hazards (Section 4). The shape is the ratio of the last forward hazard to the spot

hazard to the first deadline: rising above 1.25, front-loaded below 0.8, otherwise flat, and inverted if any forward is negative.

Calibration ratio. In a cell of observations with mids p_i and outcomes y_i, the hazard multiplier k solves

$$\sum_i [1 - (1 - p_i)^k] = \sum_i y_i. \tag{24}$$

Because $1 - (1 - p)^k = 1 - \exp(-k\,\bar{h}\tau)$, k scales every implied cumulative hazard and, under a flat hazard, is the ratio of realised to implied hazard; $k < 1$ means implied hazards are too high. Intervals are the 5th and 95th percentiles over 200 bootstrap resamples of event clusters [16]. We also report the mean of $y - p$ in cents with event-clustered t-statistics (Section 5).

Theta and event timing. Each priced observation is followed to the next grid point t_2, which is set by the grid whether or not the contract is still open ($t_2 = T$ at the 1-hour lead); the step contains a jump if $\tau_E \le t_2$. We compare the realised no-jump change with the flat-hazard theta $(1 - p_t)(1 - e^{\bar{h}_t \Delta})$, the jump count with $\sum(1 - e^{-\bar{h}_t \Delta})$, and report the martingale drift $\mathbb{E}[p_{t_2} - p_t]$, with $p_{t_2} = 1$ after a jump. Decay paths compare $p(L)/p(A)$ for contracts with no event yet at an anchor $A = 168$ hours against the flat-hazard path $[1 - (1 - p_A)^{\tau_L/\tau_A}]/p_A$. For timing, each YES contract is anchored at its earliest priced pre-event quote p_0 at t_0, with $s_0 = \max(t_0, s)$ and elapsed window share $u = (\tau_E - s_0)/(T - s_0)$. Under the flat hazard implied by p_0, $F(u) = [1 - (1 - p_0)^u]/p_0$ is uniform [24]; a mean above 0.5 means events come later than implied.

Jumps, diffusion and the Cox intensity. The quadratic variation of the hourly mid (spread at most 10 ¢) is measured from the first quote in the last 62 days to resolution and split four ways:

- diffusion: hourly changes below 20 ¢ before resolution;
- pre-resolution jumps: hourly changes of 20 ¢ or more before resolution;
- the YES resolution jump $(1 - m)^2$, from the last mid at least two hours before τ_E;
- the NO deadline step m^2.

This follows the separation of continuous and jump variation in Barndorff-Nielsen and Shephard [8] and Andersen et al. [3]. For a bounded martingale settling at 0 or 1 the expected total is $p_0(1 - p_0)$; bid-ask bounce inflates the realised value [52, 62]. On no-jump steps with mids in $(0.02, 0.98)$ we fit a CIR-type intensity [18, 26] to $\lambda = \bar{h}$ by least squares without intercept,

$$\frac{\Delta\lambda}{\sqrt{\lambda}} = \kappa\theta\,\frac{\Delta t}{\sqrt{\lambda}} - \kappa\sqrt{\lambda}\,\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon, \tag{25}$$

separately for leads above and below three days. Because $\bar{h}$ is an average hazard to T, a deadline-shaped intensity loads on κ and θ, so the fit is descriptive.

Static arbitrage and economic significance. For adjacent nested legs quoted in the same hourly candle, we count the pair-hours in which the earlier deadline's mid exceeds the later one's, and those in which its bid exceeds the later ask, before and after the taker fee on both legs. Buying YES on the later deadline and NO on the earlier one pays at least one dollar in every state; displayed sizes are not observed. The trading test is recorded briefly. A multiplier table by type, lead and price is fitted on the estimation sample, and the rule buys NO at the NO ask when the model's edge net of fees exceeds a threshold chosen on the estimation sample. It is compared with a price-only calibration model and with a control that buys NO on every quote below a dev-chosen price.

6 Results

6.1 Single-contract dynamics

We begin with the stylized facts and then document each one. Throughout, "estimation sample" means steps that end before 1 October 2026 and "holdout" means October 2026.

1. Where the sample is broad, prices behave as calibrated martingales with the fixed total variance that M0 implies.

2. The information clock depends strongly on the category. In-game sports follow the uniform clock $\lambda = 1/\tau$ almost exactly. Pre-game sports run at a few per cent of that rate, and weather information is concentrated in the last day.

3. Outside in-game sports, increments are far from Gaussian. Most hours show no mid change, and most of the variance arrives in jumps.

4. The backbone lies between Jacobi ($\beta = 1/2$) and probit. Logit-normal ($\beta = 1$) is rejected.

5. Stochastic volatility beyond the clock is present but short-lived. Filtering it improves variance forecasts under QLIKE but not under MSE.

6.1.1 Martingale property and the variance identity

For a calibrated martingale, the sum of squared price changes over a contract's life plus the final move to settlement has the same expectation as $p_0(1 - p_0)$ (Section 4). Table 16 in Appendix B reports the ratio of the two by category. Weather and sports, the two categories sampled broadly, satisfy the identity. Sampled every six hours, the ratio is 0.97 (estimation sample) and 0.93 (holdout) for weather, on 5,838 and 4,595 markets. For sports it is 0.98 and 1.01, on 9,533 and 10,007 markets. Calibration alone, $\mathbb{E}(y - p_0)^2/\mathbb{E}\, p_0(1 - p_0)$, is 0.96 and 0.94 for weather and 0.99 and 1.01 for sports. The smaller categories show ratios of 1.0–1.6. Their candle samples were partly selected by trading volume, which is ex post, so they over-represent markets that moved. We therefore do not read these ratios as evidence against the martingale.

Over six-hour horizons, the mean mid change conditional on the price bin, $\mathbb{E}[p_{t+6\mathrm{h}} - p_t \mid p_t]$, has $|t| < 2$ in 129 of 156 category-by-split-by-bin cells. The 17 cells with a mean of at least 0.3 ¢ and $|t| \geq 2$ have mixed signs and no common location in p.

6.1.2 Information clocks

Table 3 summarizes the clock fits and Figure 1 plots the piecewise information rate against the uniform clock. In-game sports are the one regime in which M0 holds in level and shape. With the uniform clock to the scheduled end, $\mathrm{Var}(u)$ is 1.02 in the estimation sample and 1.02 in the holdout, and excess kurtosis is 3–4. No alternative clock improves the holdout likelihood by more than 0.004 nats per step. This is the Brownian game model of Stern [55] observed in prices.

Pre-game sports resolve information at about 4% of the uniform rate (scaled clock, $a = 0.04$), and the rate rises steeply in the last hours before the start. Under the uniform clock $\mathrm{Var}(u)$ is 0.09 in the holdout. The bucket clock with an hour-of-day profile improves the holdout likelihood by 1.24 nats per step ($t = 8.9$). Weather concentrates information in the final day, with $\lambda \propto \tau^{-2.8}$ and an hour-of-day profile; the gain over uniform is 0.16 nats per step ($t = 10.8$).

Mentions, Politics and Entertainment track the uniform clock in level beyond about a day. On the estimation sample their gains come mainly from the hour-of-day profile: Politics gains 0.198

Table 3: Information clocks by category

	Steps	Events	Clock	Shape		ΔNLL	Var(u)		Exc. kurt.
Category	est. / holdout	est. / hold.	(BIC)	a	b	vs. uniform (t)	uniform	fitted	fitted
Sports in-game	11,536 / 9,448	5371 / 4488	Power	0.97	0.86	0.004 (5.4)	1.02	1.03	3
Sports pre-game	473,272 / 252,089	8137 / 6014	Buckets×hour	0.04	5.75	1.241 (8.9)	0.09	1.30	618
Weather	102,284 / 64,657	1189 / 840	Power×hour	0.98	2.81	0.163 (10.8)	0.88	1.03	22
Mentions	35,261 / 15,888	105 / 59	Buckets×hour	0.85	1.10	0.114 (1.9)	0.76	1.07	582
Politics	39,941 / 21,891	120 / 100	Power×hour	1.12	0.72	−0.059 (−0.3)	0.48	1.28	1281
Elections	103,634 / 0	95 / 0	Power×hour	1.13	0.98	–	1.13†	1.00†	205†
Entertainment	199,724 / 29,501	182 / 64	Buckets×hour	0.71	0.87	0.025 (0.2)	0.74	1.26	542
Sci. & Tech.	29,154 / 8,251	90 / 79	Buckets×hour	1.27	0.80	−0.093 (−1.0)	1.95	2.31	59

Hourly steps; decision quote spread $\leq 4\,¢$ and mid in $[0.03, 0.97]$. Estimation sample: steps ending before 1 October 2026; holdout: October 2026.
Clock: form with the lowest BIC on the estimation sample. *Shape*: a of the scaled clock $\lambda = a/\tau$ and b of the power clock $\lambda = a\tau^{-b}$ (uniform: $a = b = 1$).
ΔNLL: holdout mean negative log-likelihood per step of the uniform clock minus that of the chosen clock (positive: the chosen clock fits better), t clustered by event.
Var(u) and excess kurtosis of Equation (12) on the holdout. †Elections has no holdout (Section 5.1.1); estimation-sample values.

nats per step with it and 0.043 without. In Politics, Entertainment and Science & Technology the clock chosen on the estimation sample does not beat the uniform clock on the holdout. October volatility there was 1.3–2.3 times the fitted clock (Var(u) of 1.26–2.31).

6.1.3 Distribution of innovations and jumps

Figure 2 contrasts the variance and the shape of u. Once the clock is fitted, Var(u) is close to one by construction on the estimation sample. The shape, however, is far from Gaussian everywhere except in-game.

Holdout excess kurtosis is 22 for weather and 59–1281 in the other non-sports categories and pre-game sports. With the fitted clock, $P(|u| > 4)$ is 0.7–2.7% against 0.006% for a normal. Outside in-game sports (12%), the mid does not change in 22% (weather) to 85% (Elections) of hours. The EM mixture assigns 55–85% of $\sum(\Delta p)^2$ to the jump component, with similar values in the estimation sample and the holdout (Table 4). Outside in-game sports, the probability that a $|u| > 3$ hour is followed by another is 4.5–15%, against an unconditional 1.4–4.7%. Large moves cluster only mildly, so the tails are mainly isolated jumps rather than prolonged high-volatility episodes.

This matters for anyone using the model as a density. For contracts more than seven days from their scheduled end in Mentions, Politics, Elections, Entertainment and Science & Technology, the Gaussian M0 overstates the mean one-hour $|\Delta p|$ by a factor of 2.1–3.4 across price groups. The M2 mixture, which includes the zero-move mass, gives 0.7–1.2 times the realized value. Closer to expiry, and for sports and weather, the two bracket the realized move.

6.1.4 Backbone

Figure 3 plots squared price changes against the price level, after removing the clock and quote noise. Variance falls toward the edges more slowly than the probit backbone of M0 implies, and far more slowly than the logit-normal backbone $[p(1-p)]^2$. The QLIKE estimate of β is 0.39 for pre-game sports, 0.47 in-game, 0.58 for weather and 0.31–0.81 elsewhere. The Roll-corrected cell regression gives 0.44, 0.49 and 0.54 for the same three categories in the estimation sample (Table 17).

Logit-normal is rejected. It loses to probit on the estimation sample in every category (t from 3.5

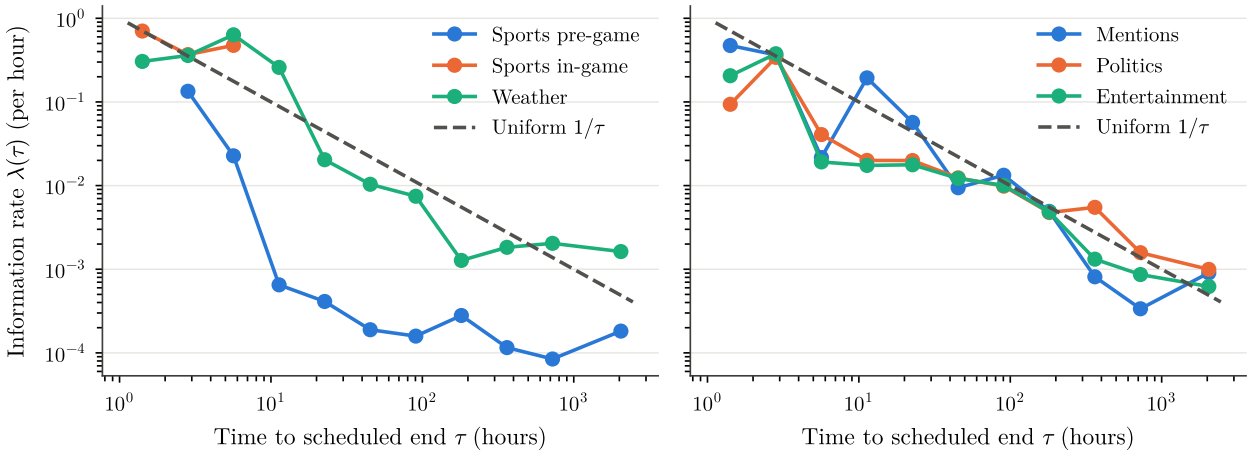


Figure 1: Information rate $\lambda(\tau)$ per hour from the piecewise-constant clock fitted on the estimation sample (before 1 October 2026), against time to the scheduled end of information. The dashed line is the uniform clock $1/\tau$; buckets without data are omitted. In-game sports sit on the uniform clock; pre-game sports run one to two orders of magnitude below it; weather information is front-loaded into the final day.

to 7.8, except Elections at 1.0). On the holdout it is worse in six of seven categories, significantly so in three (in-game $t = 4.7$, weather 2.8, Politics 2.4). Out of sample, the free-β backbone beats probit in pre-game sports ($t = -1.7$), in-game (-2.5) and weather (-1.8), and loses in Entertainment ($+1.6$).

Part of the extra volatility near the edges reflects the 1 ¢ tick: at $p = 0.04$ a one-tick move is large relative to $\varphi(\Phi^{-1}(p))$. The tick-grid control of Section 6.2.2 shows how much the grid alone distorts probit-unit statistics.

6.1.5 Stochastic volatility and leverage

Table 4 reports volatility clustering after the fitted clock is removed. The within-market first-order autocorrelation of $|u|$, which nets out static differences in volatility between markets, is 0.09–0.14 on the estimation sample (t from 2.0 to 6.4). It remains positive on the holdout, at 0.06–0.30.

The within-market half-life is 7–21 hours. The implied M1 vol-of-vol ν is 0.29–0.48 per $\sqrt{\text{day}}$. Weather is the exception, at 40 hours: its markets mostly live under 48 hours, so the within-market estimate rests on few markets. Outside weather, the pooled autocorrelation decays more slowly, with half-lives of 21–29 hours; the difference is persistent heterogeneity in α across markets.

The vol–price correlation ρ is robust only in-game. In-game, volatility rises after moves toward 1/2: the correlation is -0.11 ($t = -7.0$) on the estimation sample and -0.08 ($t = -4.2$) on the holdout in probit units, and -0.14 ($t = -8.8$) in price units. Elsewhere the probit-unit correlation is indistinguishable from zero, except Entertainment and Science & Technology ($+0.02$ and $+0.06$, $t \approx 3$). The price-unit correlation is small and negative (weather -0.03, $t = -5.7$). Because the sign depends on the units outside the in-game regime, we attribute it there to backbone and tick mechanics rather than to a structural ρ. Section 6.2 shows that the in-game effect is absent at minute scales.

6.1.6 Forecasting realized variance

Table 5 compares the forecasts on the holdout. Under QLIKE, M1 (fitted clock with the filtered vol state) beats HIST in pre-game sports (one hour: $t = -6.0$; six hours: $t = -30.3$), weather (-8.2 and

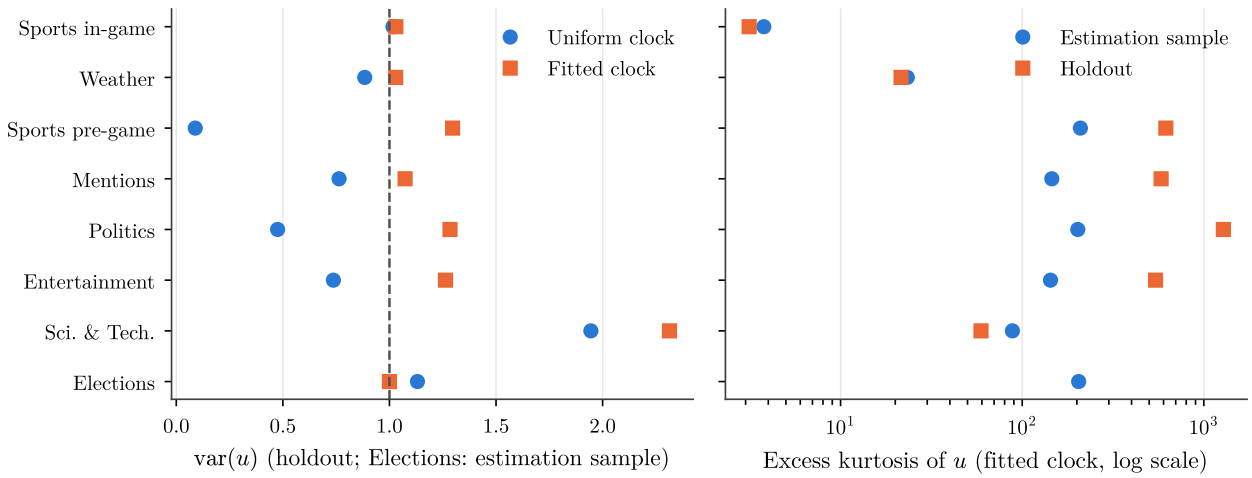


Figure 2: Standardized innovations u (Equation (12)) by category. Left: holdout (October 2026) variance under the uniform and the fitted clock; Elections has no holdout and shows the estimation sample. Right: excess kurtosis under the fitted clock, estimation sample and holdout, on a log scale. Only in-game sports are close to Gaussian.

-6.1), in-game sports (-3.0) and Politics at one hour (-2.2). The filtered state improves on M0 without it in pre-game sports ($t = -2.8$ and -2.2), in-game (-3.1), Politics at one hour (-5.2) and Mentions at six hours (-2.2). M1 does not beat HIST for Mentions, the category whose scheduled end drifts by about ten hours in October.

QLIKE and MSE disagree. Under MSE, M1 is never significantly better than M0 and is significantly worse in several cases (pre-game sports: $t = 2.3$ and 4.2). Down-weighting a quiet market pays under QLIKE, which penalizes under-prediction of small variances; MSE is dominated by the jumps that the filter cannot anticipate. Replacing the probit backbone by the fitted power backbone (M1BB against M1, holdout QLIKE) helps only for pre-game sports at one hour ($t = -2.4$). It is worse in most other cases, for example weather at one hour ($t = 5.4$).

Economic significance. Spreads that are wide relative to the model's expected move do not mean that liquidity provision earns its spread. Consider joining both sides of the best quotes every hour, with fills inferred only from trades that print through the quote, positions held to settlement, and Kalshi maker fees. This loses 1.25 ¢ per filled contract on the estimation sample ($t = -3.7$, 27,036 fills) and 1.56 ¢ on the holdout ($t = -4.0$, 17,055 fills). Restricting quotes to hours in which the half-spread is at least 20 times the model's expected one-hour move, a threshold chosen on the estimation sample, gives +1.61 ¢ per fill on the holdout ($t = 0.7$, 350 fills). That is no better than a plain half-spread filter of at least 10 ¢ with no model (+1.70 ¢, $t = 0.9$). The cost to a liquidity provider is adverse selection on the fills that do trade through, not diffusion risk, which the clock measures well. We do not pursue trading applications further.

Limitations.

- The candle samples of the smaller categories were selected partly on volume, which affects the variance-identity ratios and limits the number of events.
- Elections has no holdout because its October series have no prior schedule.
- The Mentions schedule estimated on the estimation sample is about ten hours off in October.

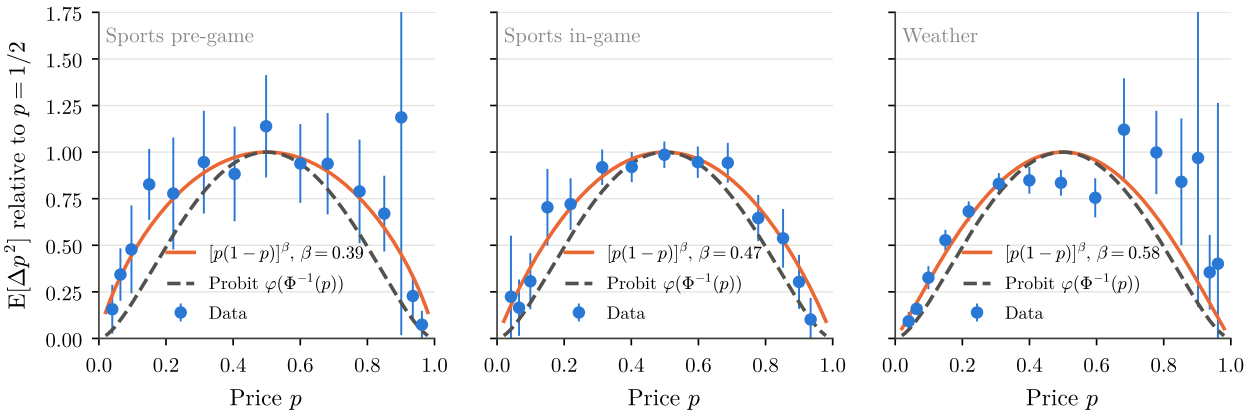


Figure 3: Backbone. Points: mean of $[(\Delta p)^2 - ks^2]/c^2_{j(\tau)}$ by price bin, estimation sample (before 1 October 2026), with ±2 event-clustered standard errors; c_j and k are from the fitted power backbone, so the series is normalized to one at $p = 1/2$. The solid line is the fitted $[p(1-p)]^{2\beta}$, normalized the same way; the dashed line is the probit backbone $\varphi(\Phi^{-1}(p))^2$ of M0. The logit-normal curve ($\beta = 1$) lies below both and is omitted.

- Hourly candles cannot separate jumps from short bursts of stochastic volatility; the in-game L2 data in Section 6.2 address this for sports.

6.2 High-frequency in-game dynamics

The in-game sample has 50 game-winner legs (NBA 14, NCAAF 14, NFL 10, NHL 12), with 6,965 in-game minutes over two nights (Section 5.1.5). Every statistic is reported next to the matched M0 control of Section 5.1.6, which has the same games, windows, starting prices and spreads but no jumps and no stochastic volatility. Five facts emerge.

1. The variance scales like Brownian motion from one second to five minutes, at a level set by the league.
2. The mid shows no measurable bid–ask bounce, unlike a quote re-rounded to the tick each second.
3. The shape is far from Gaussian at seconds and becomes nearly Gaussian only at about five minutes, because about half of the variance arrives in jumps.
4. Jump sizes follow Stern's fixed-score scaling in basketball but not in football or hockey.
5. There is no vol–price correlation beyond what M0 implies at minute scales.

6.2.1 Scale invariance and the variance signature

Table 6 reports the standardized innovation under the uniform clock at each horizon. Variance is flat in the horizon. $\mathrm{Var}(u)$ is 0.71, 0.77, 0.77, 0.73 and 0.77 at 1, 5, 15, 60 and 300 seconds, so the Brownian time scaling holds across two and a half orders of magnitude. The level is below one (t of $u^2 - 1$ between −3.5 and −5.1, by game). With start spreads of at most 2 ¢ it is 0.66–0.77.

The level is set by the league, and the per-league values are also flat in the horizon. NBA games run above the uniform clock ($\mathrm{Var}(u)$ of 1.17–1.33); NCAAF (0.47–0.54), NFL (0.57–0.64) and NHL (0.57–0.67) run below it. A plausible reading is that the scheduled end is a poor proxy for the

Table 4: Stochastic volatility, leverage and jumps

	Corr($\lvert u_t\rvert, \lvert u_{t+1}\rvert$)			Half-life	ν	Leverage corr. (t)		Jump share	Zero-
Category	pooled	within (t)	within, hold.	(h)	per $\sqrt{\text{day}}$	probit	price	est. / hold.	move
Sports in-game	0.12	–	0.22	–	–	−0.109 (−7.0)	−0.144 (−8.8)	0.66 / 0.65	0.12
Sports pre-game	0.16	0.14 (6.4)	0.21	9.4	0.48	0.001 (0.2)	−0.009 (−2.2)	0.74 / 0.72	0.73
Weather	0.13	0.12 (2.0)	0.30	40.1	0.16	−0.010 (−1.7)	−0.034 (−5.7)	0.62 / 0.55	0.22
Mentions	0.09	0.09 (5.5)	0.06	6.8	0.44	0.006 (0.6)	−0.003 (−0.3)	0.85 / 0.82	0.72
Politics	0.11	0.09 (4.6)	0.28	11.6	0.33	0.009 (0.7)	−0.031 (−2.4)	0.78 / 0.85	0.76
Elections	0.13	0.13 (3.5)	–	21.2	0.29	−0.001 (−0.1)	−0.023 (−1.5)	0.81 / –	0.85
Entertainment	0.11	0.09 (5.9)	0.12	13.4	0.33	0.022 (3.2)	0.006 (0.8)	0.70 / 0.73	0.81
Sci. & Tech.	0.14	0.10 (4.4)	0.24	11.6	0.40	0.059 (3.5)	0.021 (1.5)	0.76 / 0.80	0.72

Hourly u under the fitted clock. Estimation sample: before 1 October 2026; holdout: October 2026. Unless marked, statistics are from the estimation sample.
Within: $|u|$ demeaned by market, markets with at least 48 steps (in-game legs are too short). Half-life and ν come from the within-market fit $A\phi^k$, $k = 1, \ldots, 24$ (Section 5.1.3).
Leverage: $\text{Corr}(\text{sgn}(Z_t)x_t, \frac{1}{6}\sum_{j\le 6}|x_{t+j}|)$ for $p \in [0.1, 0.9]$, with $x = u$ (probit) or $x = \Delta p$ over the backbone volatility (price); negative means volatility rises after moves toward 1/2.
Jump share: posterior-weighted share of $\sum(\Delta p)^2$ from the EM jump component (Equation (15)). *Zero-move*: share of hours without a mid change.
t clustered by event.

remaining game clock: long stoppages in football and the buffer in the scheduled expiration both make τ too long.

The shape is not scale-invariant. Excess kurtosis falls from 151 at one second to 51, 24 and 9 at 5, 15 and 60 seconds, and to 2 at five minutes; $P(|u| > 4)$ falls from 0.99% to zero. The control behaves differently. In the matched control the tick grid inflates $\text{Var}(u)$ to 2.64 at one second, falling to 1.02 at one minute, and kurtosis stays below 7.

Figure 4 shows the variance signature. Realized variance of the mid per in-game hour is flat, between 453 and 524 c^2 for sampling intervals from 1 to 600 seconds. Fitting Equation (16) gives $\sigma^2 = 503\,\text{c}^2$ per hour and a noise variance at its lower bound of zero. The control has the textbook signature: 1,148 c^2 at one second, falling to 608 at one minute, with $\eta = 0.27\,¢$ and noise accounting for 46% of one-second realized variance. Kalshi in-game mids therefore do not flicker across the tick. The book moves when the price moves, and quote discreteness appears as stickiness rather than bounce. The slight rise of realized variance from 1 to 10 seconds (453 to 524 c^2) is mild positive autocorrelation at the seconds scale: quotes take a few seconds to adjust. Beyond about ten seconds increments are uncorrelated. For realized-variance estimation this means sparse sampling corrections [1, 35, 62] are unnecessary on this venue at these horizons.

6.2.2 The probit-drift shortfall

In the hourly data the outward probit drift, which keeps p a martingale under M0, is only a fraction of its predicted size. The instrumental-variable estimates of Section 5.1.2 are no closer to one, so quote noise in Z_t does not explain it. Table 7 and Figure 5 test whether the price grid does. In weather, Mentions, Elections and Science & Technology, M0 paths quoted on the 1 ¢ grid with the real spreads reproduce the measured slope (data 0.57, 0.42, 0.35, 0.47; tick control 0.32, 0.48, 0.37, 0.53). There the shortfall comes from the grid, the 0.5 ¢ probit clip and, for weather, the steep clock, which together truncate outward moves near the edges. Ticks explain it only in part for Entertainment (0.35 against 0.71) and Politics (0.73 against 1.03).

They do not explain it for sports: pre-game hourly 0.11 (s.e. = 0.08) against 0.93, and in-game L2 0.46 (0.21) at 60 seconds and 0.40 (0.25) at 300 seconds, against 1.37 and 0.87. Stale quotes add

Table 5: Forecasting next-period realized variance (holdout)

				QLIKE difference (t)			MSE rel. to HIST		MSE
Category	H	Rows	Events	M1 − HIST	M1 − M0	M0 − HIST	M0	M1	M1−M0 (t)
Sports in-game	1	5,685	2568	−0.43 (−3.0)	−0.03 (−3.1)	−0.40 (−2.7)	0.32	0.32	(1.2)
Sports pre-game	1	208,130	3204	−0.58 (−6.0)	−0.56 (−2.8)	−0.01 (−0.1)	1.19	1.40	(2.3)
Weather	1	10,746	770	−0.36 (−8.2)	−0.11 (−1.9)	−0.26 (−3.4)	0.42	0.42	(0.5)
Mentions	1	13,781	36	0.96 (1.1)	−0.36 (−1.4)	1.33 (1.5)	0.96	0.97	(0.1)
Politics	1	16,106	83	−0.74 (−2.2)	−0.39 (−5.2)	−0.35 (−1.1)	0.57	0.79	(1.7)
Entertainment	1	28,394	56	−3.27 (−1.6)	−0.07 (−0.2)	−3.20 (−1.3)	0.94	0.97	(1.4)
Sci. & Tech.	1	6,961	75	−0.28 (−2.1)	0.05 (0.4)	−0.34 (−2.3)	0.10	1.02	(2.8)
Sports pre-game	6	194,028	2854	−2.54 (−30.3)	−0.41 (−2.2)	−2.13 (−8.5)	0.40	0.85	(4.2)
Weather	6	3,873	546	−0.48 (−6.1)	−0.19 (−0.9)	−0.29 (−1.1)	0.33	0.34	(3.3)
Mentions	6	12,071	34	0.07 (0.3)	−0.24 (−2.2)	0.31 (0.9)	0.50	0.56	(1.2)
Politics	6	15,446	51	−0.70 (−1.6)	−0.23 (−1.0)	−0.47 (−1.2)	0.05	0.69	(1.9)
Entertainment	6	27,379	54	−4.45 (−1.5)	−0.15 (−0.7)	−4.30 (−1.3)	0.81	1.01	(1.0)
Sci. & Tech.	6	5,451	65	−0.08 (−0.5)	0.01 (0.1)	−0.09 (−0.6)	0.03	0.57	(2.5)

Target: $RV_{t,H} = \sum_{j<H} (\Delta p_{t+j})^2$ (cents^2), H in hours; decision rows with at least 24 h of history. All forecasts, windows, EWMA half-lives and weights are calibrated on the estimation sample (before 1 October 2026); the table reports the holdout (October 2026). Elections has no holdout.

QLIKE differences are mean per-row loss differences (negative: the first model is better), t clustered by event. MSE columns are holdout MSE relative to HIST; the last column is the t-statistic of the per-row MSE difference M1 − M0 (positive: M1 worse).

nothing. Re-quoting only in hours with real activity gives the same slopes as the tick control.

Three further results locate the residual shortfall and rule out under-reaction as its source. The price itself does not drift. The mean of $\mathrm{sgn}(Z_t)\Delta p$ is $+0.003$ ¢ per hour pre-game ($t = 1.3$) and -0.004 ¢ per minute in-game ($t = -0.1$). A Gaussian M0 price with the measured Z-drift would drift by -0.020 ¢ per hour pre-game, which lies outside the confidence band of the measured value. The shortfall is not in the tails. Trimming with windows that bias in opposite directions leaves the slope low: pre-game 0.13 and 0.05, in-game 0.68 and 0.61 at 60 seconds. The core carries little of the clock. Within the core ($|u| \le 2$), $\mathbb{E}(\Delta Z)^2$ is only 8% (pre-game), 19% (Mentions, Entertainment) and 34% (in-game, 60 s) of the clock increment L. Within that core, the local second-order martingale ratio $\mathbb{E}[\mathrm{sgn}(Z)\Delta Z]/\mathbb{E}[\frac{1}{2}|Z|(\Delta Z)^2]$ is 1.35–2.26.

So the diffusive part of the motion drifts outward at least as much as a martingale requires given its own variance. The shortfall arises because most of the clock's information arrives in jumps, and the Gaussian compensator $e^{L/2}$ of M0 overstates the drift needed when the variance is jump-borne. That is the M2 regime of Section 4, not under-reaction. The only under-reaction we find is the seconds-scale adjustment visible in the signature plot.

6.2.3 Jumps and Stern's scaling

Panel A of Table 8 measures how much of the in-game variance arrives in jumps. Bipower variation misses 59%, 53% and 48% of realized variance at 5, 10 and 15 seconds; the median game shows 61–66%. In the control, bipower variation misses only 8–16%: the familiar small-sample and discreteness bias of bipower tests on a tick grid. The excess over the control, about 40–45 percentage points, is a conservative estimate of the in-game jump share. Threshold jumps (more than four local standard deviations and at least 2 ¢) carry 51–68% of realized variance, against 1% or less in the control. At 10 seconds we detect 859 jumps in 50 games, about 7.4 per in-game hour, with a median size of 3.5 ¢. The control produces 58. These shares are consistent with the hourly EM

Table 6: In-game M0 test by horizon: data and matched tick-grid control

		Data					M0 control (1 ¢ grid)		
h (s)	Steps	Var(u)	(t)	Exc. kurt.	$P(\lvert u\rvert > 4)$, %	Var(u), 2 ¢	Var(u)	Exc. kurt.	$P(\lvert u\rvert > 4)$, %
1	319,821	0.71	(−5.1)	151.3	0.99	0.66	2.64	6.3	2.97
5	63,909	0.77	(−3.6)	51.1	0.99	0.74	1.42	1.1	0.47
15	21,277	0.77	(−3.6)	23.7	0.79	0.76	1.16	0.2	0.06
60	5,265	0.73	(−4.4)	9.0	0.53	0.73	1.02	0.1	0.00
300	991	0.77	(−3.5)	2.0	0.00	0.77	1.07	0.0	0.00
By league (data):		Var(u)	*at h = 1, 5, 15, 60, 300 s; excess kurtosis at 1, 60, 300 s*						
NBA (14 games)		1.17	1.33	1.31	1.26	1.29	62.6	3.2	0.2
NCAAF (14)		0.47	0.48	0.49	0.48	0.54	161.3	9.9	1.8
NFL (10)		0.57	0.61	0.64	0.62	0.59	167.4	7.6	2.6
NHL (12)		0.65	0.65	0.65	0.57	0.67	309.9	24.1	4.4

Recorded Kalshi L2, nights of 7–8 and 8–9 November 2026; one game-winner leg per game, 50 games. Non-overlapping h-second steps under the uniform clock to the scheduled end; start spread ≤ 3 ¢ ("2 ¢": ≤ 2 ¢), mid in $[0.03, 0.97]$; end spread ≤ 10 ¢.
(t): t-statistic of $\mathbb{E}[u^2 - 1]$, clustered by game. The control simulates an M0 path per game and quotes it on the 1 ¢ grid with that game's spreads (Section 5.1.6).
In the league rows, the columns under Data hold Var(u) at the five horizons and the control columns hold the data's excess kurtosis at 1, 60 and 300 s.

estimate for in-game sports (0.65–0.66 of $\sum(\Delta p)^2$, Table 4).

For a fixed score increment, Stern's Brownian game model implies that the price jump scales as $\varphi(Z)/\sqrt{\tau}$ (Panel B). In the NBA the log–log slope is 1.08 (s.e. = 0.09), as predicted.

For the NFL, NCAAF and NHL the slope is 0.50 (0.10), 0.56 (0.08) and 0.65 (0.16); the pooled slope with league fixed effects is 0.69 (0.07).

Basketball scoring is frequent and small, close to the Brownian limit. Touchdowns, field goals and goals are large and discrete, and their price impact depends on the game state through more than $\varphi(Z)$. Measuring τ to the scheduled end rather than the game clock also flattens the slope.

6.2.4 Leverage at minute scales

At horizons of one, two and five minutes (Panel C), the vol–price correlation in probit units is +0.016, +0.003 and +0.045, all with $|t| < 1.4$, matching the control's zero. In price units the correlation is negative (−0.056 to −0.088). M0 itself produces this mechanically, because $\varphi(Z)$ rises as the price moves toward 1/2; the control gives −0.07 to −0.17. The hourly in-game correlation of −0.11 in probit units (Table 4) therefore has no counterpart at minute scales. If it is structural, it reflects game-state changes over hours, such as the late stages of close games, rather than short-run feedback.

Limitations.

- The sample covers two nights and 50 games, so league-level estimates rest on 10–14 games each.
- The in-game window comes from the schedule and can include a few pre-kickoff minutes, which pushes Var(u) down.
- τ is measured to the scheduled settlement time, not the game clock.
- Without a score feed, "jumps" are price discontinuities, not verified scoring events.

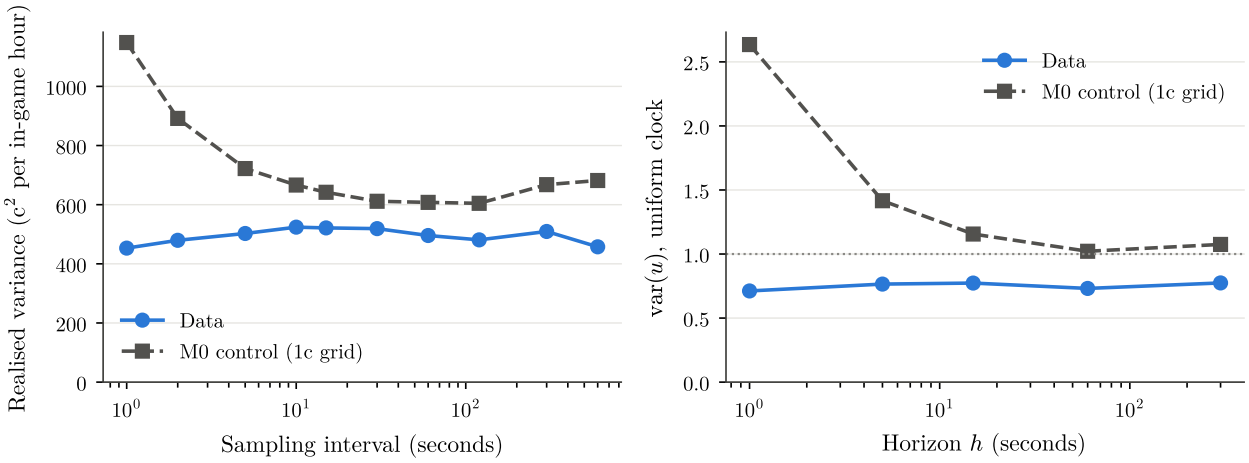


Figure 4: In-game variance signature and time scaling, recorded L2, two nights, 50 games. Left: realized variance of the mid per in-game hour against the sampling interval. Right: Var(u) under the uniform clock against the horizon; the dotted line marks one. Dashed: the matched M0 control, quoted on the 1 ¢ grid with each game's own spreads.

6.3 Strike ladders: the prediction-market smile

Sample. The fitted sample has 24,192 snapshots of 10,177 ladders in 9,213 events from 309 series, priced from 232,594 leg quotes. There are 12,627 snapshots in the estimation sample and 11,565 in the holdout. Sports ladders account for 16,342 snapshots and weather for 7,271. Politics (270 snapshots), entertainment (207), science and technology (57) and other climate series (44) are small. Table 18 in Appendix C gives coverage by group. Most attrition comes from legs without a recent candle or with a spread above 5 ¢. Among weather lows, where the middle brackets are often quoted wider than 5 ¢, it also comes from incomplete bracket sets.

6.3.1 The shape of the smile

Every group has a smile (Table 9 and Figure 6): the curvature c in (20) is positive, with a 95% interval excluding zero, in every group, lead and sample. Implied wings are therefore fatter than a normal distribution with the ATM width. The skew differs by underlying.

- *Sports totals and team totals* are right-skewed, $s = 0.071$ and 0.046 six hours before expiry. Upper-wing volatilities exceed lower-wing ones, as for a count bounded below.
- *Spreads*, oriented to the favourite, are nearly symmetric ($s = 0.010$).
- *Weather highs and lows* carry a small left skew ($s = -0.017$ and -0.031).
- *Politics* ladders have no significant skew and a curvature of about 0.07, at the upper end of the range (weather lows and entertainment reach 0.087 and 0.088).

The pooled curvature at a given lead is stable across leads: weather highs give $c = 0.054$–0.055 at every lead from 36 to 16 hours (Table 19). The median per-snapshot curvature is smaller than the pooled one for sports, for example -0.005 against 0.014 for totals at six hours. The pooled curvature is driven by the far wings, where (Figure 6) the volatility ratio rises steeply beyond $|m| \approx 2$. Part of that rise is mechanical: a wing quoted at the 1 ¢ tick has an implied volatility that grows with $|m|$ whatever the true tail.

Table 7: Probit-drift slope: data, tick-grid controls and where the shortfall lies

	Slope (M0 = 1)				Trimmed slope	Core	Local	$\mathbb{E}[\mathrm{sgn}(Z)\Delta p]$
Sample	Data (s.e.)	Tick (s.e.)	Stale	True	$\lvert u\rvert \le 2$ / $\lvert\Delta Z\rvert \le 2$sd	$\mathbb{E}(\Delta Z)^2/\mathbb{E}L$	ratio	¢ (t)
Weather (1 h)	0.57 (0.05)	0.32 (0.01)	0.28	0.57	0.69 / 0.62	0.49	1.35	−0.021 (−1.0)
Mentions (1 h)	0.42 (0.12)	0.48 (0.10)	0.43	0.88	0.39 / 0.37	0.19	1.86	0.026 (1.5)
Elections (1 h)	0.35 (0.10)	0.37 (0.12)	0.34	0.84	–	–	–	0.006 (1.6)
Sci. & Tech. (1 h)	0.47 (0.16)	0.53 (0.10)	0.41	0.69	–	–	–	0.026 (1.8)
Entertainment (1 h)	0.35 (0.15)	0.71 (0.10)	0.71	1.16	0.25 / 0.25	0.19	1.82	−0.003 (−0.9)
Politics (1 h)	0.73 (0.11)	1.03 (0.21)	0.99	1.11	0.58 / 0.56	0.22	2.19	−0.010 (−0.9)
Sports pre-game (1 h)	0.11 (0.08)	0.93 (0.09)	0.92	1.06	0.13 / 0.05	0.08	1.43	0.003 (1.3)
In-game L2 (60 s)	0.46 (0.21)	1.37 (0.39)	–	1.29	0.68 / 0.61	0.34	2.26	−0.004 (−0.1)
In-game L2 (300 s)	0.40 (0.25)	0.87 (0.32)	–	0.93	0.51 / 0.47	0.48	1.35	0.031 (0.1)

Slope of $\tilde{Z}_{t+h} - \tilde{Z}_t$ on $\tilde{Z}_t(e^{L/2} - 1)$, steps with $L \le 2$. Hourly rows: estimation sample (before 1 October 2026), category clock fitted on the estimation sample. L2 rows: in-game, uniform clock. Standard errors clustered by event (hourly) or game (L2).
Tick, *Stale*, *True*: M0 paths simulated on the same grids, quoted on the 1 ¢ grid with the real spreads, re-quoted only in hours with real activity, or not quoted (Section 5.1.6).
Trimmed: windows centred on M0's mean (biases toward 1) and on zero drift (biases toward 0). *Core* and *Local ratio*: within $\lvert u\rvert \le 2$; local ratio = $\mathbb{E}[\mathrm{sgn}(Z)\Delta Z]/\mathbb{E}[\frac{1}{2}\lvert Z\rvert(\Delta Z)^2]$ on $\lvert Z\rvert \ge 0.5$ (one for a martingale). The trimming diagnostics were run for the categories shown.
Last column: per hour (hourly rows) or per step (L2 rows), t clustered as above.

The skew-t summaries compare like with like (Table 9, last four columns). The implied skew has the right sign and roughly the right size. For sports totals the implied Bowley skewness is 0.123 and 0.101 in the two samples, against 0.116 and 0.067 realised. Weather highs are left-skewed in both the implied and the realised fits. Implied tails, however, are consistently heavier than realised ones. The implied tail weight is 1.13–1.40 for sports and 1.23–1.29 for weather highs, against 1.03–1.19 and 1.04–1.05 realised. That gap is the cross-strike form of the favourite–longshot bias [32, 54, 58].

6.3.2 Smile dynamics: sticky moneyness, not sticky strike

Table 10 reports the regressions (21)–(22) on consecutive snapshot pairs. Two pieces of evidence reject a sticky-strike smile.

The ATM point does not slide along the smile. When the median moves, σ_{ATM} barely responds. The slope b is 0.017 for sports totals, −0.031 for team totals, −0.020 for spreads and −0.040 for weather highs. A smile fixed in strike would give the pooled skew instead (0.065, 0.039, 0.013 and −0.014). Using each snapshot's own skew, the coefficient δ that should equal 1 under sticky strike is between 0.06 and 0.54 in the weather and sports groups, and $\delta = 1$ is rejected at $\lvert t\rvert \ge 4.1$ for totals, team totals and weather highs.

The volatility at a fixed strike moves with the median (Figure 7). With the pooled smile, the sticky-moneyness coefficient γ is 1.01 for spreads, 1.17 for weather highs, 1.31 for totals and 2.11 for team totals (t from 9 to 25 against zero). Sticky strike ($\gamma = 0$) is rejected in every group. For totals and team totals the coefficient also exceeds one: the smile moves more than one-for-one with the median, possibly because the wings re-price as the game state changes.

The raw response has the sign sticky moneyness predicts. In sports totals, where $s > 0$, the volatility at a fixed strike falls when the median rises: −0.081 per unit of ΔF_n ($t = -15.9$). With each snapshot's own smile the coefficients fall to 0.25–0.63 in sports, consistent with attenuation from measurement error in snapshot-level (s, c). Weather highs, whose snapshot smiles are better determined, give 1.17 under either definition.

Backbone. For sports totals, σ_{ATM} rises with the level of the median both across games ($\beta = 0.29$,

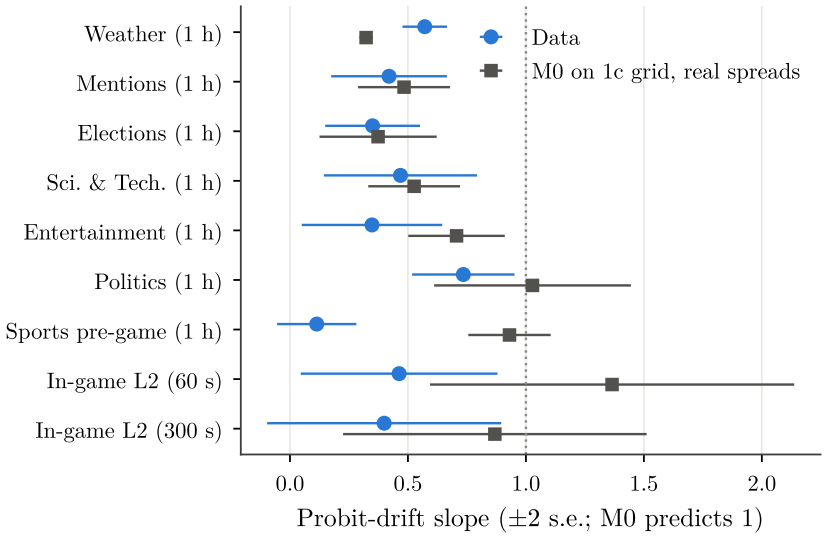


Figure 5: Probit-drift slope (±2 standard errors, clustered by event or game) in the data and in M0 paths quoted on the 1 ¢ grid with the real spreads. Hourly rows: estimation sample (before 1 October 2026). L2 rows: in-game, two nights. The dotted line is M0's prediction. Ticks reproduce the shortfall in weather, Mentions, Elections and Science & Technology but not in sports.

$t = 13.2$) and within a game between snapshots ($\beta = 0.28$, $t = 3.2$). Both are well below the square-root exponent $\frac{1}{2}$ of a Poisson count. For team totals the cross-section gives $\beta = 0.22$ but the within-game exponent is slightly negative (-0.12). The SABR-style level backbone is therefore weak: the volatility of a ladder is set mainly by its moneyness and its clock, not by the level of the underlying.

6.3.3 ATM term structure

Figure 8 plots $\log \sigma_{\text{ATM}}$ against the lead within ladders. The estimated exponents η (Table 20) separate three information clocks.

- *Weather highs* have $\eta = 0.773$ [0.760, 0.786], well above the uniform-clock value $\frac{1}{2}$. Uncertainty about the daily maximum resolves late, during the afternoon of the measurement day.
- *Weather lows* have $\eta = 0.480$ [0.413, 0.546], indistinguishable from $\frac{1}{2}$. Their profile is flat from 16 to 8 hours, after the morning minimum has been observed.
- *Politics, entertainment and science* ladders are sub-diffusive (0.35, 0.19 and 0.24). Their information arrives in lumps close to resolution.
- *Sports ladders* have no pre-game time decay at all ($\eta = 0.001$–0.009): nothing is learned about the score before the game starts.

6.3.4 Calibration against the realised outcome

Table 11 and Figure 9 test the implied distribution against X_T. Sports ladders are calibrated in width and shape. In the holdout the width ratio is 1.02–1.03 (no $|t|$ above 1.1). The fractional PIT histogram is flat, with outer-decile shares of 0.192–0.202 against 0.2. In the estimation sample the

Table 8: In-game jumps, Stern scaling and leverage

A. Share of realized variance from jumps					
h (s)	Bipower	Bipower, med. game	Bipower, control	Threshold	Threshold, control
5	0.59	0.66	0.16	0.68	0.005
10	0.53	0.61	0.09	0.57	0.009
15	0.48	0.63	0.08	0.51	0.010
B. Jump size against $\varphi(Z)/\sqrt{\tau}$, 10 s jumps (Stern: slope 1)					
League	Jumps	Games	Med. $\lvert\Delta p\rvert$ (¢)	Price slope (s.e.)	Probit slope (s.e.)
Pooled (league f.e.)	859	50	3.5	0.69 (0.07)	0.72 (0.10)
NBA	236	14	4.0	1.08 (0.09)	1.92 (0.12)
NCAAF	187	14	3.5	0.56 (0.08)	0.41 (0.14)
NFL	284	10	3.5	0.50 (0.10)	0.55 (0.11)
NHL	152	12	3.0	0.65 (0.16)	0.59 (0.21)
C. Leverage, Equation (14) with x for u (t)					
h (s)	Steps	$x = u$, data	$x = u$, control	$x = \Delta p$, data	$x = \Delta p$, control
60	4,447	0.016 (0.9)	0.000 (0.0)	−0.056 (−2.3)	−0.068 (−2.9)
120	2,181	0.003 (0.1)	−0.007 (−0.3)	−0.088 (−2.8)	−0.106 (−3.5)
300	822	0.045 (1.4)	0.008 (0.2)	−0.075 (−1.7)	−0.165 (−4.7)

Recorded L2, two nights, 50 games; steps with start spread ≤ 3 ¢ and mid in $[0.03, 0.97]$ (Panel C: $[0.1, 0.9]$). Control: M0 paths quoted on the 1 ¢ grid with each game's spreads.
A: bipower share $= (RV - BV)/RV$; threshold jumps have $|u| > 4$ local bipower standard deviations and $|\Delta p| \geq 2$ ¢. B: price slope of $\log|\Delta p|$ on $\log[\varphi(Z)/\sqrt{\tau}]$; probit slope of $\log|\Delta Z|$ on $\log \tau^{-1/2}$; OLS, standard errors clustered by game. C: correlation of $\operatorname{sgn}(Z_t)x_t$ with the mean $|x|$ over the next five steps; t clustered by game; negative values mean volatility rises after moves toward 1/2.

widths are slightly too narrow (totals 1.045, $t = 2.2$; spreads 1.144, $t = 4.0$), and spreads put too much mass in the outer deciles (0.234); neither carries over to the holdout.

Weather is too wide. The width ratio for highs is 0.941 ($t = -2.4$) and 0.890 ($t = -3.8$), and the PIT puts too little mass in the outer deciles for both highs and lows (0.168–0.180, t from −3.2 to −6.3). Sports totals show a location bias: the realised total lands above the implied median by $0.121\,\sigma_{\text{ATM}}$ in the estimation sample ($t = 5.1$) and $0.064\,\sigma_{\text{ATM}}$ in the holdout ($t = 2.4$).

The tail-mass comparison (Figure 9, right; Table 21) locates the excess width in the wings.

- *Weather highs.* Beyond $|m| \geq 2.5$ the market prices 0.93–1.11 ¢ of tail mass per strike; realised frequencies are 0–0.30 ¢ (z_{H_0} between −0.5 and −2.6, all negative).
- *Weather lows.* Every wing and threshold has $z_{H_0} < 0$.
- *Sports.* The evidence is mixed. Holdout upper tails of totals at $|m| \geq 2.5$ are overpriced (1.70 ¢ implied, 0.11 ¢ realised, $z_{H_0} = -2.2$), but the same cell in the estimation sample is not (1.77 against 2.19 ¢).
- *Politics and entertainment.* Beyond $|m| \geq 1.5$, wings are priced at 0.8–3.3 ¢ and are realised less often in most cells, but with at most 64 events per cell no z_{H_0} is below −1.3.

6.3.5 Implied versus realised variance between snapshots

Figure 10 and Table 22 report the ratio (23). *Sports, pre-game.* Both terms are nearly zero, since neither the median nor the width moves, and the ratio is not informative. *Sports, in play* (6 hours to 1 hour before expiry). Realised variance falls short of the implied variance drop:

Table 9: Implied smile and density shape by ladder group.

		Smile, (20)		Bowley skewness		Tail weight	
Group	Lead	s (t)	c (t)	Estimation	Holdout	Estimation	Holdout
Weather highs	24 h	−0.017 (−14.4)	0.055 (62.2)	−0.015 / −0.077	−0.020 / −0.033	1.290 / 1.050	1.231 / 1.044
Weather lows	36 h	−0.031 (−5.3)	0.067 (18.1)	–	–	–	–
Sports totals	6 h	0.071 (66.3)	0.014 (12.6)	0.123 / 0.116	0.101 / 0.067	1.184 / 1.082	1.200 / 1.030
Team totals	6 h	0.046 (30.0)	0.023 (18.7)	0.095 / 0.070	0.054 / 0.028	1.395 / 1.075	1.245 / 1.057
Spreads	6 h	0.010 (4.4)	0.041 (8.9)	0.035 / 0.035	0.002 / 0.026	1.364 / 1.190	1.129 / 1.055
Politics	6 h	0.008 (0.8)	0.070 (18.1)	0.008 / 0.074	0.031 / −0.036	2.588 / 1.422	2.605 / 1.161

Smile: pooled least squares of $\sigma_K/\sigma_{\text{ATM}} - 1$ on $(1, m, m^2)$ over off-ATM strikes at the lead shown, estimation sample (before 1 October 2026) and holdout (October 2026) pooled; t clustered by event. All leads, samples and 95% intervals are in Table 19. Bowley skewness and tail weight (1 for a normal) come from Jones–Faddy skew-t fits in standardised units over all leads; each cell is implied / realised. Implied fits use the pooled survival points; realised fits are interval-censored maximum likelihood on $(X_T - F)/\sigma_{\text{ATM}}$. Cluster-bootstrap 90% intervals of the implied tail weight are at most 0.08 wide in the sports and weather-high cells. Weather-low skew-t fits are unstable (implied tail weight 2.59 in the estimation sample and 1.02 in the holdout) and are omitted.

Table 10: Smile dynamics between consecutive snapshots.

		ATM, (21)		Fixed strike, (22)			Backbone β	
Group	Pairs	b (t)	$\bar{s}$	γ pooled ($t_{\gamma=1}$)	γ snapshot	$\partial\sigma_K/\partial F$ (t)	Within (t)	Cross (t)
Weather highs	4,230	−0.040 (−7.4)	−0.014	1.17 (3.6)	1.17	0.019 (2.0)	–	–
Sports totals	3,984	0.017 (1.4)	0.065	1.31 (4.5)	0.63	−0.081 (−15.9)	0.28 (3.2)	0.29 (13.2)
Team totals	2,566	−0.031 (−2.8)	0.039	2.11 (10.1)	0.30	−0.117 (−14.1)	−0.12 (−2.9)	0.22 (13.8)
Spreads	2,250	−0.020 (−1.5)	0.013	1.01 (0.1)	0.25	−0.003 (−0.4)	–	–
Politics	155	−0.056 (−0.8)	0.017	0.68 (−2.5)	0.33	0.055 (1.8)	0.69 (1.2)	0.20 (1.9)

Pairs of consecutive snapshots of one ladder, estimation sample (before 1 October 2026) and holdout (October 2026) pooled. Fixed effects for each group and lead pair; t clustered by event. b: slope of $\Delta\sigma_{\text{ATM}}/\sigma_{\text{ATM}}$ on $\Delta F/\sigma_{\text{ATM}}$. $\bar{s}$: mean pooled skew, the sticky-strike prediction for b (sticky moneyness predicts 0). The coefficient δ on $s\,\Delta F/\sigma_{\text{ATM}}$ with each snapshot's skew (1 under sticky strike) is 0.23, 0.06, 0.12, 0.54 and −0.31 in the five rows ($t_{\delta=1} = -4.9$, −4.1, −5.3, −1.3, −6.4). γ: coefficient on the sticky-moneyness prediction $-(s + 2cm)\Delta F/\sigma_{\text{ATM}}$, using the group-and-lead pooled smile or each snapshot's own smile (1 under sticky moneyness, 0 under sticky strike); the pooled γ has t between 9.4 and 24.6 against zero (5.3 for politics). Fixed-strike pairs: 5,259 (weather highs), 28,108 (totals), 16,490 (team totals), 24,867 (spreads), 730 (politics). $\partial\sigma_K/\partial F$: raw slope of the fixed-strike change on $\Delta F/\sigma_{\text{ATM}}$. Backbone: $\Delta \log \sigma_{\text{ATM}}$ on $\Delta \log F$ within ladders, and $\log \sigma_{\text{ATM}}$ on $\log F$ with series-by-lead fixed effects, for positive underlyings only.

- totals 0.90 [0.81, 0.98];
- spreads 0.92 [0.84, 1.04];
- team totals 0.65 [0.59, 0.70].

The implied distribution narrows faster than its median moves: a variance premium, sizeable only for team totals.

Weather highs change sign across the day. The ratio is 1.02 from 36 to 24 hours, 0.66 [0.58, 0.73] overnight (24 to 16 hours) and 1.08 in the morning (16 to 12 hours). It reaches 1.55 [1.37, 1.74] in the afternoon (12 to 8 hours), when the median moves more than the narrowing of the distribution allows. The weather median is also not a martingale. Its standardised drift is −0.046 overnight ($t = -4.5$) and positive in the other three windows (t from 2.8 to 5.8): the market under-forecasts the daytime maximum until observations arrive.

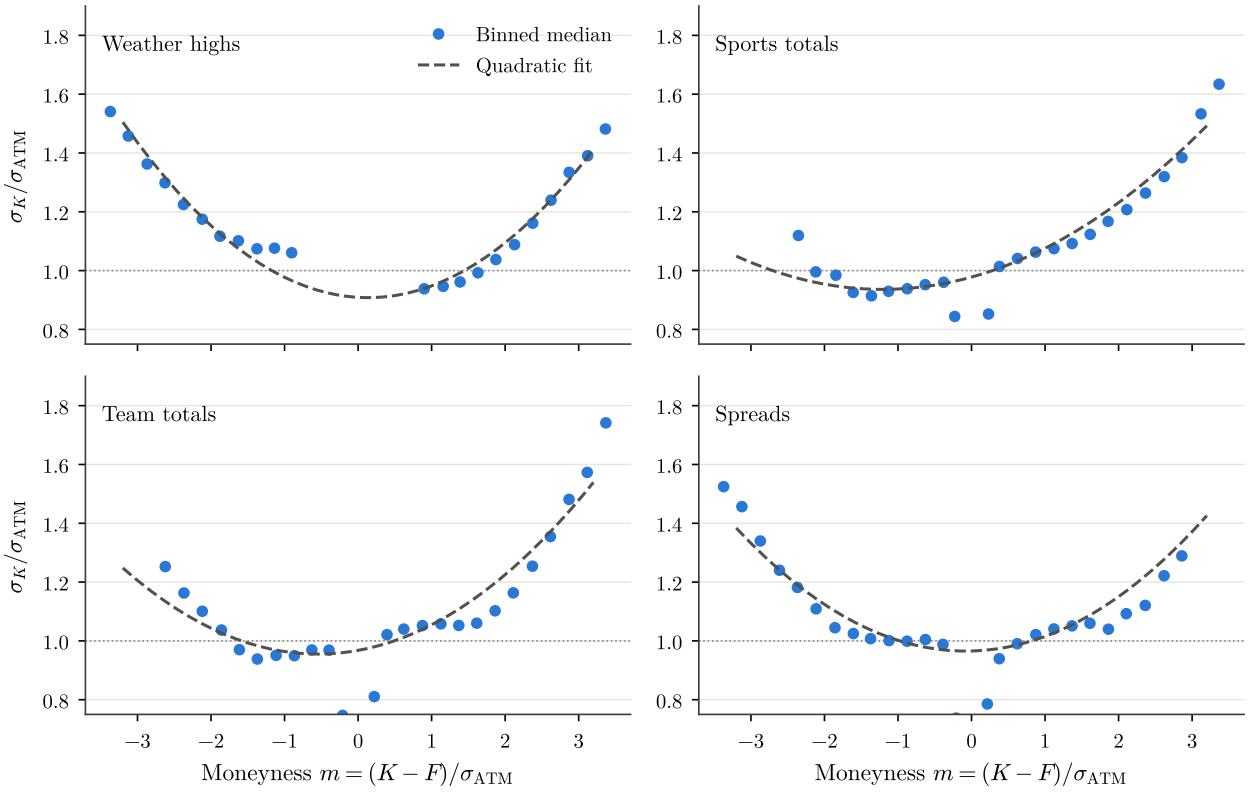


Figure 6: Implied volatility smile. Median $\sigma_K/\sigma_{\mathrm{ATM}}$ in moneyness bins of width 0.25 (bins with at least 30 strikes), all leads and both samples pooled, with the pooled quadratic fit of (20). The dotted line is the flat smile of a normal implied distribution. Bins near $m = 0$ are noisy because $\sigma_K = (F - K)/\Phi^{-1}(S)$ has a small denominator there.

6.3.6 Economic significance

The wing premium is small in money terms.

- *Gating by model moneyness.* Selling the tail side of every leg with model moneyness $|m| \geq 2.5$ at the snapshot quote, after the taker fee, returns $+0.23\,¢$ per contract in the estimation sample ($t_{\mathrm{risk}} = 0.4$, 2,741 contracts) and $+1.10\,¢$ in the holdout ($t_{\mathrm{risk}} = 2.0$, 3,006 contracts).
- *Gating by price alone.* The control (tail price at most $2\,¢$) returns $+0.31$ and $+0.25\,¢$.
- *Within one price band.* Among tails priced at most $5\,¢$, legs with $|m| \geq 2$ return $+0.29$ and $+0.49\,¢$ while legs with $|m| < 2$ return -1.22 and $-0.85\,¢$: moneyness separates the overpriced wings from the rest.

The gain per contract is bounded by the $1\,¢$ tick, and the evidence is weak once settlement risk is taken into account. Net-of-fee arbitrage between legs (monotonicity or bracket sums) appears in 30 of 24,192 snapshots.

6.3.7 Limitations

Several limitations apply.

- *Quotes.* They are hourly candle closes up to two hours old, without displayed size, and closes of different legs are not synchronous.

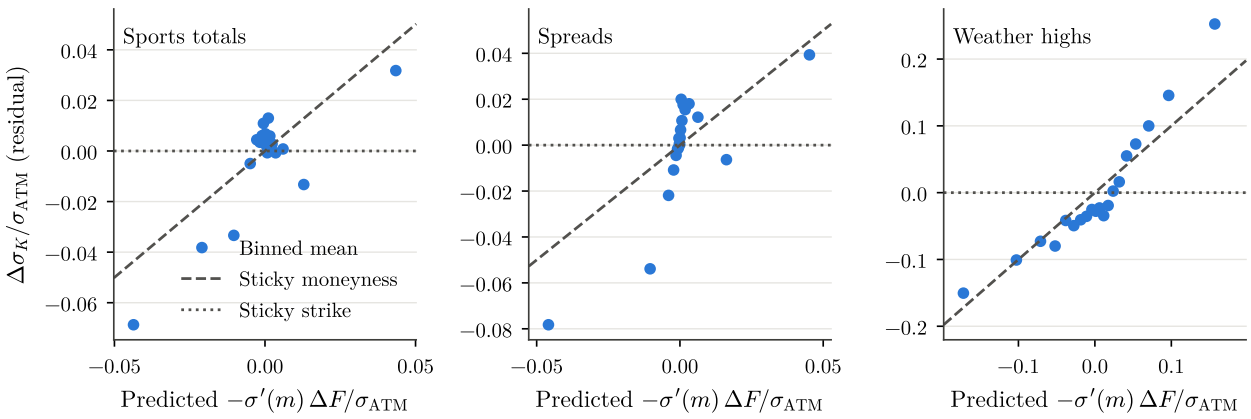


Figure 7: Fixed-strike volatility change against the sticky-moneyness prediction $-\sigma'(m)\,\Delta F/\sigma_{\mathrm{ATM}}$ (pooled smile), after removing group-by-lead-pair means and partialling out the level term of (22). Points are means in 20 quantile bins of the prediction. Sticky moneyness predicts the dashed 45-degree line, sticky strike the dotted zero line. In sports most pairs have small median moves; the slope is identified by the larger moves.

Table 11: Calibration of the implied distribution against the realised outcome.

	Width ratio λ ($t_{\lambda=1}$)		Location μ (t)		PIT outer-decile share ($t_{0.2}$)	
Group	Estimation	Holdout	Estimation	Holdout	Estimation	Holdout
Weather highs	0.941 (−2.4)	0.890 (−3.8)	0.066 (1.9)	−0.095 (−2.4)	0.178 (−3.2)	0.170 (−3.9)
Weather lows	0.916 (−0.7)	0.949 (−0.7)	−0.081 (−1.5)	0.064 (0.8)	0.168 (−6.3)	0.180 (−3.5)
Sports totals	1.045 (2.2)	1.021 (1.0)	0.121 (5.1)	0.064 (2.4)	0.196 (−0.6)	0.192 (−0.9)
Team totals	1.045 (1.6)	1.025 (1.0)	0.127 (3.7)	0.021 (0.6)	0.199 (−0.1)	0.202 (0.2)
Spreads	1.144 (4.0)	1.027 (1.1)	0.000 (0.0)	−0.047 (−1.5)	0.234 (2.7)	0.201 (0.1)
Politics	1.216 (1.0)	1.297 (1.8)	−0.152 (−0.9)	0.052 (0.3)	0.192 (−0.2)	0.233 (0.9)
Entertainment	1.179 (1.9)	0.852 (−1.6)	0.013 (0.1)	0.056 (0.3)	0.205 (0.2)	0.164 (−2.8)

One observation per snapshot, all leads pooled; estimation sample (before 1 October 2026), holdout (October 2026). λ, μ: interval-censored normal maximum likelihood for $(X_T - F)/\sigma_{\mathrm{ATM}}$; $\lambda < 1$ means the implied ATM width is too wide. PIT: fractional histogram of the interval-censored probability integral transform; share of mass in the deciles $[0, 0.1)$ and $[0.9, 1]$, 0.2 under calibration. All t statistics cluster by event. Spread locations are in the favourite's orientation.

- *Spread filter.* The 5 ¢ filter discards many weather-low bracket sets and most far wings that have no bid.
- *Tick floor.* Wings quoted at the 1 ¢ tick inflate far-wing volatilities, so part of the measured curvature is a price-grid effect.
- *Interval censoring.* The realised outcome is known only up to the bracket containing it. This limits the precision of the realised skew-t fits, which carry no intervals.
- *Fixed-strike coefficient.* With snapshot smiles it is attenuated by measurement error. With pooled smiles it inherits any misspecification of a common smile within a group and lead.
- *Sports clock.* The expected expiration of sports markets is a scheduled time, so the 1 hour snapshot falls at different stages of different games.
- *Multiple comparisons.* We report many groups, leads and tests. The statements above rely on patterns that recur across groups or samples rather than on single cells.

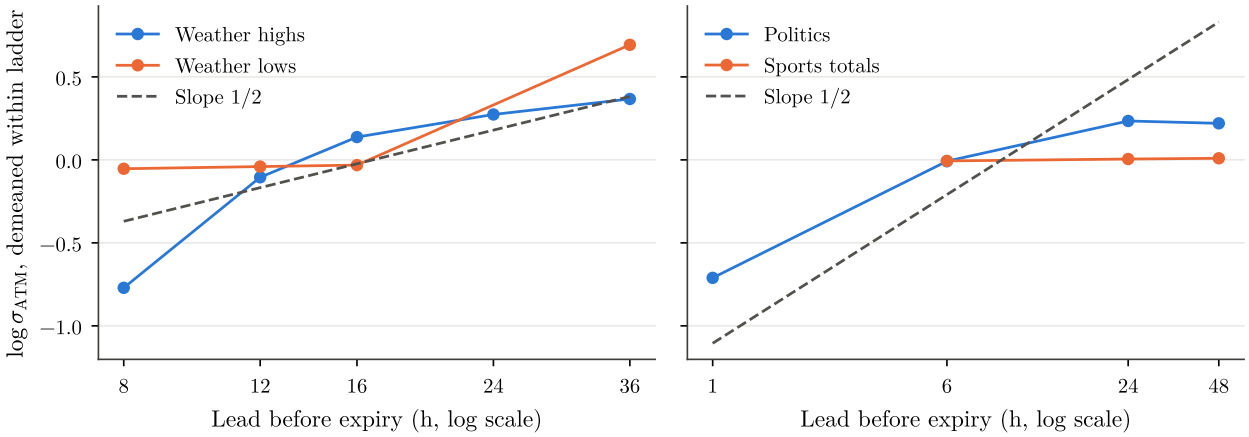


Figure 8: ATM term structure. Mean of $\log \sigma_{\text{ATM}}$, demeaned within ladder, by lead before expiry (log scale). Left: weather highs and lows (hours before the measurement-day close). Right: politics and sports totals (hours before the expected expiration; sports pre-game leads only). The dashed line has slope $\frac{1}{2}$, the uniform information clock.

6.4 Deadline contracts: hazard term structures and theta

6.4.1 Implied hazards are too high, most of all in the last day

Table 12 reports the hazard multiplier k of Equation (24) for the pooled event, mention and count contracts, by lead and split. In the estimation sample the implied hazard is too high at every lead. Realised hazards are about 61% of implied ones within a day of the deadline and 61–79% at longer leads, and every 90% interval lies below one. In cents, prices exceed realised frequencies by 1.2–8.6 ¢ per contract, and the gap grows with the lead because longer windows carry higher prices. The holdout confirms the near-deadline result: $k = 0.62$ within six hours and 0.51 at 6–24 hours, with intervals below one. Beyond three days the holdout point estimates lie between 0.69 and 0.98, but with 23–52 clusters per cell the intervals include one.

Figure 11 splits the pooled multiplier by contract type. One-off events carry the largest overpricing near the deadline, $k = 0.47$ [0.34, 0.62] within six hours, and lie between 0.58 and 0.76 at longer leads. Mentions are overpriced within a day ($k \approx 0.65$) but close to calibrated beyond a week, where the intervals include one. Eliminations, whose resolution is a scheduled episode, show $k \approx 0.68$–0.70 within a day and are calibrated beyond three days. By category, the strongest near-deadline overpricing is in science and technology ($k = 0.23$ [0.12, 0.36] within a day) and politics (0.50 [0.40, 0.62]). Elections, sports beyond a week, and climate are not distinguishable from calibrated (Table 24).

The longshot cells carry most of the near-deadline effect in relative terms. Within six hours, quotes with mids below 5 ¢ average 1.3 ¢, yet only 0.1% of them resolve YES. For 5–15 ¢ quotes, k ranges from 0.34 to 0.82 across leads (Table 25). Part of the sub-5 ¢ gap is mechanical. With a one-cent tick the mid cannot fall below half a cent, and a one-cent ask is the cheapest YES on offer. The pattern matches the time-to-expiry deterioration of calibration documented by Page and Clemen [49] and the longshot overpricing surveyed by Snowberg and Wolfers [54]. Here it appears in hazard units and concentrates in the final day.

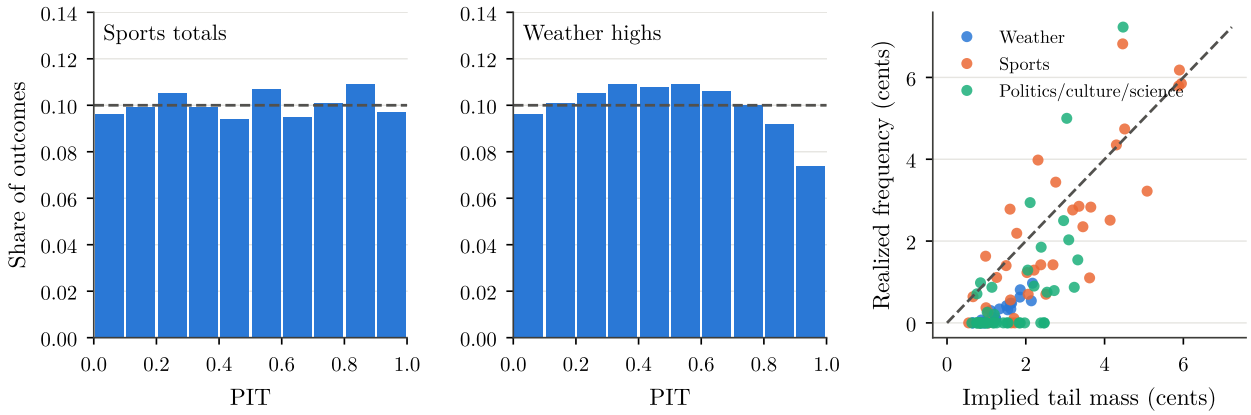


Figure 9: Calibration. Left and centre: fractional PIT histograms in the holdout (October 2026) for sports totals and weather highs; the dashed line is the uniform density. Right: implied tail mass against realised frequency, per group, sample, wing and threshold $|m| \geq 1, \ldots, 2.5$ (thresholds from 1.5 shown), in cents; the dashed line is calibration.

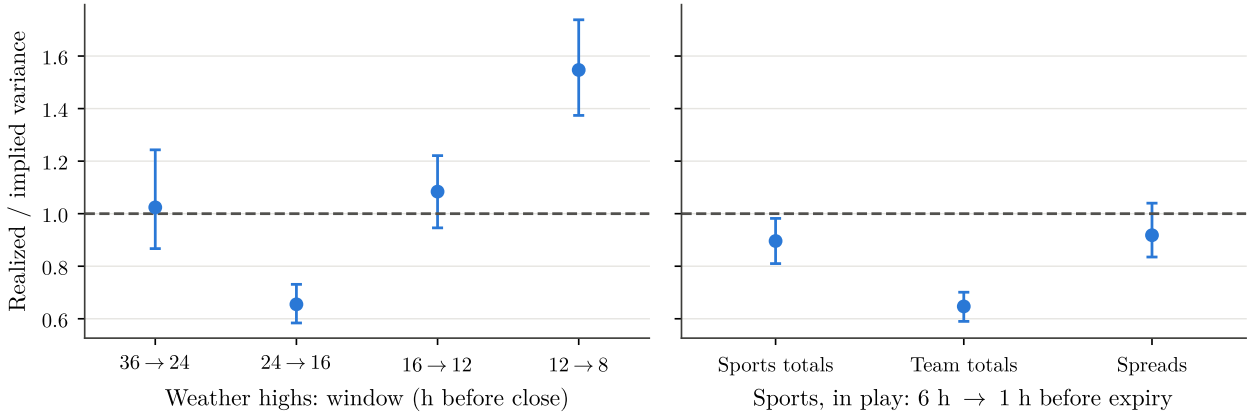


Figure 10: Realised over implied variance between consecutive snapshots, (23), with 90% cluster-bootstrap intervals over events. Left: weather highs by window (hours before the measurement-day close). Right: sports ladders in play, 6 to 1 hours before the expected expiration. The dashed line is equality.

6.4.2 Implied hazard term structures

Across strips, the forward hazard curve is more often rising than flat. Of 3,019 daily snapshots on 194 strips, 48% are rising, 35% front-loaded, 15% flat and 2.4% inverted, and the median ratio of the last forward hazard to the spot hazard is 1.20 (Table 13). The shape differs by category in an interpretable way:

- Science and technology strips, mostly product-release dates, are rising in 61% of snapshots with a median ratio of 2.0. The market expects a release later rather than sooner.
- Climate strips (storm and seasonal milestones) are also rising.
- Entertainment, sports and elections lean front-loaded (median ratios 0.70–0.89): if the occurrence does not come soon, it becomes less likely.
- Politics is mixed, with a median ratio of 1.10.

Table 12: Calibration of implied hazards by lead (events, mentions and counts)

	Estimation sample (before 1 October 2026)				Holdout (October 2026)			
Lead	n (cl.)	Mid	k [90% int.]	$\bar{y} - \bar{p}$ (t)	n (cl.)	Mid	k [90% int.]	$\bar{y} - \bar{p}$ (t)
< 6 h	3,658 (438)	0.143	0.61 [0.55, 0.68]	−1.23 (−7.30)	376 (66)	0.117	0.62 [0.44, 0.92]	−1.35 (−1.90)
6–24 h	3,236 (335)	0.197	0.61 [0.55, 0.68]	−2.25 (−6.77)	393 (60)	0.142	0.51 [0.38, 0.67]	−2.98 (−2.94)
1–3 d	3,058 (315)	0.243	0.73 [0.65, 0.82]	−2.27 (−4.19)	357 (43)	0.243	0.71 [0.50, 0.93]	−2.46 (−1.62)
3–7 d	3,300 (321)	0.336	0.79 [0.72, 0.88]	−3.22 (−3.83)	351 (52)	0.304	0.98 [0.71, 1.35]	−0.23 (−0.09)
7–14 d	1,396 (249)	0.193	0.68 [0.56, 0.80]	−4.14 (−3.96)	97 (42)	0.250	0.69 [0.45, 1.24]	−4.38 (−1.23)
14–30 d	1,301 (256)	0.303	0.68 [0.59, 0.78]	−5.93 (−4.59)	118 (42)	0.424	0.84 [0.54, 1.19]	−2.58 (−0.66)
30–60 d	968 (199)	0.346	0.61 [0.52, 0.71]	−8.55 (−5.51)	48 (23)	0.587	0.85 [0.52, 1.44]	−2.44 (−0.52)

One observation per contract and grid lead with spread at most 10¢; n observations and, in parentheses, event clusters. Mid is the mean quote mid. k solves Equation (24) (realised/implied cumulative hazard; $k < 1$: implied hazard too high). Its interval is the 5th–95th percentile over 200 bootstrap resamples of event clusters. $\bar{y} - \bar{p}$ is the mean outcome minus mid in cents (¢), with the t-statistic clustered by event. The estimation sample contains contracts with deadlines before 1 October 2026; the holdout contains decisions taken in October 2026.

Table 13: Hazard term-structure shapes and static-arbitrage frequency by category

	Forward-hazard shape (daily snapshots)						Adjacent nested pairs (hourly)			
Category	Snap.	Strips	$h_{\text{fwd}}/h_{\text{spot}}$	Rising	Front	Inv.	Pair-h	$p_1 > p_2$	Crossed	Exec.
Politics	1,631	102	1.10	0.44	0.35	0.024	63,520	0.022	0.0016	0.0008
Science & tech.	783	38	2.02	0.61	0.29	0.022	32,955	0.033	0.0012	0.0007
Entertainment	400	22	0.89	0.41	0.45	0.022	15,108	0.060	0.0017	0.0012
Sports	126	17	0.70	0.43	0.48	0.048	14,173	0.044	0.0018	0.0011
Elections	46	9	0.89	0.39	0.46	0.022	2,196	0.010	0.0005	0.0005
Climate & weather	33	6	1.74	0.70	0.09	0.000	2,467	0.149	0.0004	0.0004
All	3,019	194	1.20	0.48	0.35	0.024	130,419	0.034		

Left: one snapshot per strip and day (16:00 UTC candle, spread at most 10¢), legs open, windows started and nested. $h_{\text{fwd}}/h_{\text{spot}}$ is the median ratio of the last forward hazard to the spot hazard to the first deadline. Rising: ratio above 1.25; Front: below 0.8; Inv.: some forward hazard negative; the remaining share is flat. Right: adjacent nested legs quoted in the same hourly candle. $p_1 > p_2$: the earlier deadline's mid exceeds the later one's. Crossed: the earlier bid exceeds the later ask. Exec.: the crossing survives the taker fee on both legs. Shares of pair-hours; all splits pooled.

Along a single contract, the implied hazard rises steeply as the deadline approaches without an event (Figure 12a). Relative to its value seven days out, the median $\bar{h}$ is 1.4 times higher at 72 hours, 2.3 times at 24 hours, 7 times at 6 hours and 33 times at one hour for politics. Science and technology and entertainment follow nearly the same path; mentions are milder, at 10 times at one hour. A flat deterministic hazard would keep this ratio at one. Taken at face value, prices imply a hazard concentrated near the deadline; the timing evidence below shows that realised hazards are concentrated there too, but by much less.

6.4.3 Theta: prices decay too slowly near the deadline

Conditional on no event, prices fall more slowly than the flat-hazard theta at every lead (Table 14). The realised no-event change is 28–69% of the model change, and the gap is significant everywhere (t between 3.6 and 11.5). Jumps are also rarer than the implied hazard predicts: 18 against 74 expected in the last six hours, and 40 against 98 at 6–24 hours. The two effects net out to a

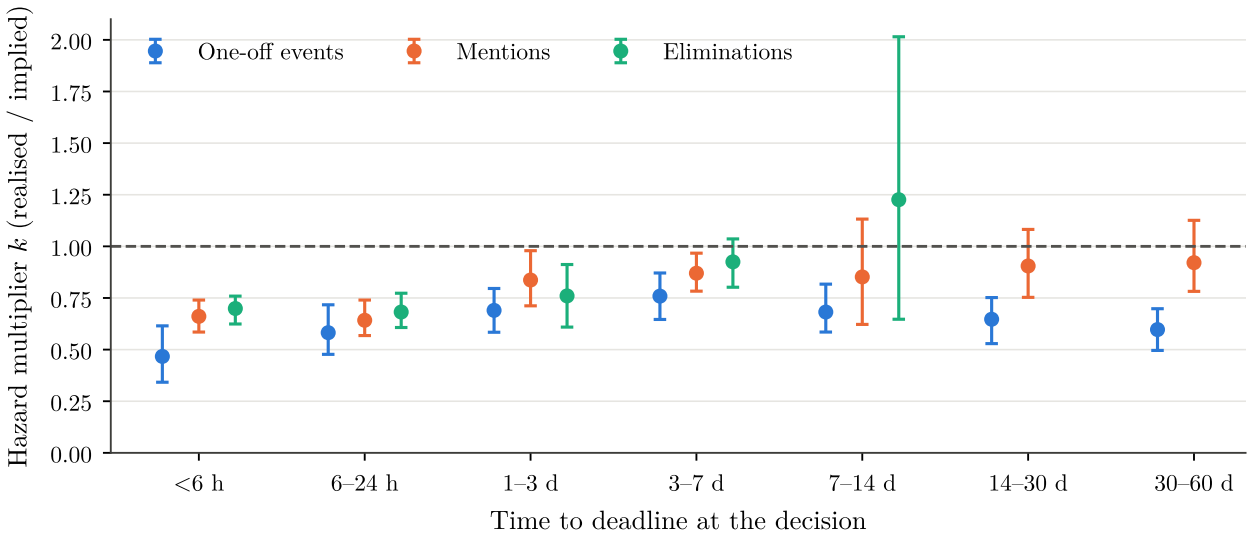


Figure 11: Hazard multiplier k (realised over implied cumulative hazard) by lead and contract type, all splits pooled, with 5th–95th percentile intervals from 200 cluster-bootstrap resamples. The dashed line is perfect calibration. Cells with fewer than 30 observations are omitted.

near-zero martingale drift at most leads. The exception is the last six hours, where prices fall by 0.86 ¢ per grid step on average ($t = -7.2$), and marginally the 14–30 day leads (-1.3 ¢, $t = -2.1$). Read together, slow decay and rare jumps mean that the price path is close to a martingale away from the deadline but not in the final hours. In those hours the excess is not removed by decay and is paid off only at settlement.

Figure 12b follows the 15–50 ¢ cohort from seven days out. With no event, the mean price is still 41% of its starting level 24 hours before the deadline (the flat model gives 16%), 28% at six hours (model 4%) and 15% one hour before (model 0.7%). In levels this is 10.4 ¢ against 4.2 ¢ at 24 hours and 3.2 ¢ against 0.2 ¢ at one hour. Longshots starting below 15 ¢ (mean 5.6–6.2 ¢) settle on a plateau of 1.4–2.1 ¢ inside the last day, against model values of 0.03–0.9 ¢. This plateau is largely the tick floor. Eliminations show the scheduled-event version of the same picture: no-event decay is only 12–31% of flat theta beyond six hours, because the price should not decay until the episode airs.

6.4.4 When do deadline events happen?

Resolutions arrive later in the window than the flat implied hazard predicts (Figure 13a). For the 807 event, mention and count contracts that resolved YES (238 clusters), the mean probability integral transform is 0.61 ($t = 7.7$; Kolmogorov–Smirnov distance 0.16). Of these resolutions, 15.7% fall in the last tenth of the window against 6.4% under the flat hazard, and 20% in the final day. One-off events are the most back-loaded (mean PIT 0.65, $t = 8.3$; 19.7% in the last tenth). Mentions follow at 0.60 ($t = 4.6$), while counts are consistent with a flat hazard (0.52, $t = 0.35$). Eliminations are a scheduled reveal: their PIT is 0.95 and 84% of resolutions fall in the last tenth.

This evidence and the calibration results fit together. Realised hazards do rise towards the deadline, which is why a flat hazard misplaces events. But the market's implied hazard rises much faster (Figure 12a). In the final six hours, the realised number of resolutions is a quarter of what the prevailing implied hazards predict (18 against 74, Table 14). The net result is a deadline premium concentrated in the final day.

Table 14: Theta, jumps and martingale drift over the next grid step (events, mentions and counts)

			Jumps		No-event change (¢)			
Lead	n (cl.)	Drift ¢ (t)	Real.	Flat	Real.	Flat	Ratio (t gap)	$\sigma_{\ln \bar{h}}$
$<$ 6 h	3,580 (451)	−0.86 (−7.18)	18	74.2	−1.19	−1.85	0.65 (10.1)	2.23
6–24 h	3,108 (355)	−0.35 (−1.55)	40	98.1	−1.14	−2.40	0.48 (10.2)	1.22
1–3 d	2,771 (317)	0.34 (1.12)	72	168.9	−1.23	−4.43	0.28 (11.5)	0.79
3–7 d	2,994 (321)	−0.31 (−0.58)	243	341.3	−4.42	−6.40	0.69 (3.6)	0.53
7–14 d	1,303 (240)	−0.28 (−0.59)	27	82.2	−1.38	−3.98	0.35 (5.6)	0.39
14–30 d	1,188 (238)	−1.30 (−2.10)	65	126.1	−3.76	−6.00	0.63 (4.5)	0.27
30–60 d	792 (189)	−0.28 (−0.32)	70	106.0	−3.83	−6.38	0.60 (3.6)	0.21

Priced observations (spread at most 10¢, mid below 0.9, no event yet), all splits pooled; the step runs to the next grid point (to T at the 1-hour lead). Drift: mean price change with the price set to 1 after a jump, t clustered by event. Jumps: realised count against the flat-hazard expectation $\sum(1 - e^{-\bar{h}\Delta})$. No-event change: mean realised mid change against the flat-hazard theta $(1 - p)(1 - e^{\bar{h}\Delta})$. Ratio: realised over model, with the clustered t of their difference. $\sigma_{\ln \bar{h}}$: standard deviation of $\Delta \ln \bar{h}/\sqrt{\Delta}$ per square-root day on no-jump steps. Conditioning on "no event" uses the event-time upper bound and is descriptive.

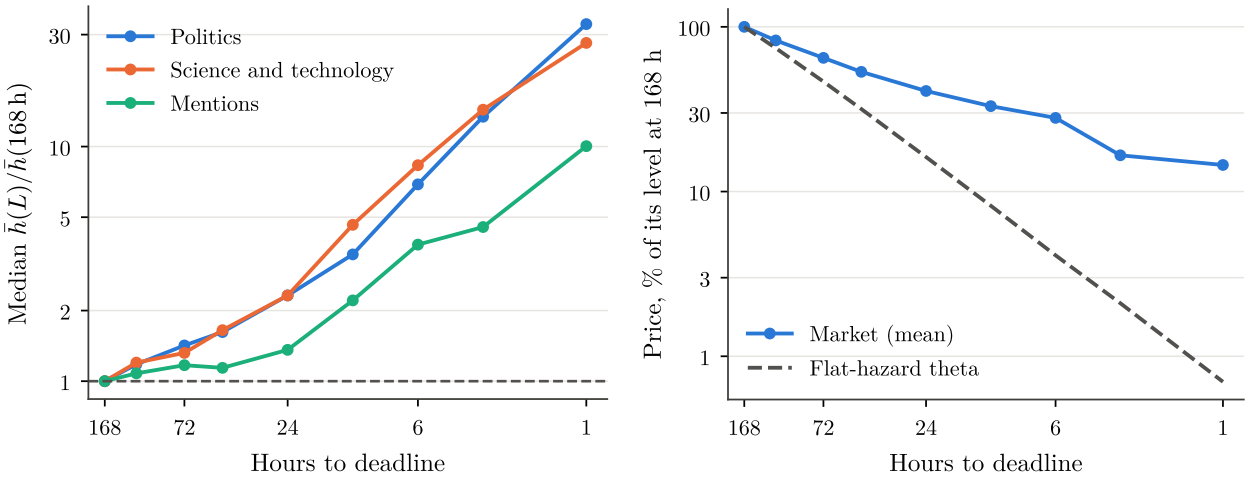
6.4.5 Jumps, diffusion and the implied-hazard process

Most price variance before resolution is news, not the resolution itself (Figure 13b). For the 4,032 event, mention and count contracts with enough hourly quotes, hourly changes below 20 ¢ account for 34% of quadratic variation, and jumps of at least 20 ¢ before resolution (about 0.5 per contract) for 42%. The YES resolution jump accounts for 23%, and the NO deadline step for 0.8%. Prices of contracts that resolve YES stand at 84 ¢ on average two hours before the event-time bound, so the final jump averages 16 ¢. The total is 1.19 times $\sum p_0(1 - p_0)$, close to the martingale identity, with the excess consistent with bid-ask bounce. Among events, mentions and counts, the YES resolution share is smallest for one-off events (16%) and largest for counts (37%). Eliminations are dominated by pre-resolution jumps (69%), as episodes reveal information in large steps.

The implied hazard itself diffuses, with a volatility that rises towards the deadline. The standard deviation of $\Delta \ln \bar{h}$ is 0.21–0.39 per square-root day beyond a week and 2.2 within six hours (Table 14). A CIR fit (Equation (25)) is rejected as a description of $\bar{h}$: the estimated mean reversion is negative at every horizon, $\kappa = -0.05$ per day beyond three days and -2.0 within three days. The implied average hazard drifts upward towards the deadline instead of reverting (Table 28). A Cox model for these contracts therefore needs a deterministic deadline-shaped component in the intensity, with diffusion around it of roughly 0.4 per square-root day in log terms far from the deadline.

6.4.6 Static arbitrage across strips

Monotonicity violations are common at the mid but almost never executable (Table 13). Over 130,419 pair-hours on 898 adjacent nested pairs, the earlier deadline's mid exceeds the later one's 3.4% of the time, but the earlier bid exceeds the later ask in only 0.04–0.18% of pair-hours by category. After taker fees, 108 pair-hours on 52 pairs in 36 strips remain, with a median locked-in payoff of 0.8 ¢ per pair at unknown displayed size. Calendar spreads show that forward segments are overpriced as well. In the estimation sample, the implied conditional probability of an event between the two deadlines is 0.27 against a realised 0.21, and buying the forward (NO on the near leg, YES on the far leg) loses 8.5 ¢ per pair ($t = -5.0$). In the holdout it loses 7.2 ¢ ($t = -2.2$; 105 observations on 31 pairs).



(a) Median implied hazard relative to 168 h

(b) Theta path, 15–50¢ cohort

Figure 12: Implied hazard and price paths of deadline contracts with no event yet, anchored seven days (168 h) before the deadline. (a) Median of $\bar{h}(L)/\bar{h}(168\,\mathrm{h})$ for the three largest categories; a flat deterministic hazard gives one (dashed). (b) Mean price relative to its anchor value for contracts priced 15–50¢ at the anchor, against the flat-hazard theta path $[1-(1-p_A)^{\tau_L/\tau_A}]/p_A$ (dashed). Both axes are logarithmic. Conditioning on no event uses the event-time upper bound and is descriptive.

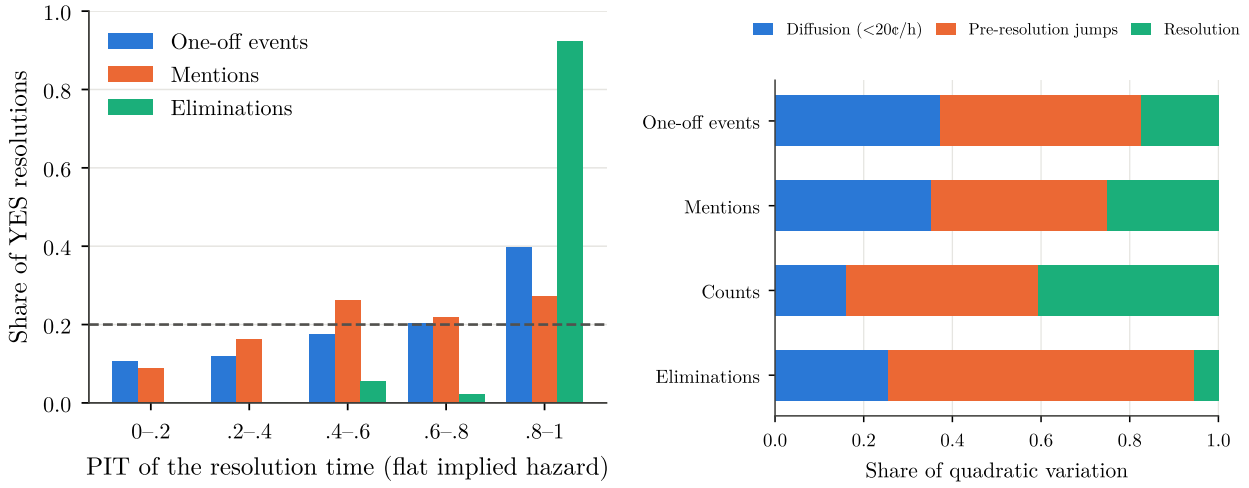
6.4.7 Economic significance

The overpricing does not yield a robust trading rule (Table 29). Buying NO when the dev-fitted multiplier table shows an edge earns 19.0 ¢ per contract in-sample but −5.3 ¢ ($t = -0.8$; 49 trades in 13 clusters) in the holdout. A price-only calibration model does no better. Buying NO on every quote below 50 ¢ earns 1.3 ¢ in the holdout ($t = 1.65$), and 0.06 ¢ on the first signal per contract. In a logistic blend with the mid, the hazard model's holdout coefficient is 0.006 ($t = 0.04$): out of sample it adds no information to the price.

6.4.8 Limitations

Several limitations qualify these results:

- The holdout is short: 2,076 priced observations, with at most 66 event clusters per lead cell.
- The event time is an upper bound. It makes resolutions look slightly late, biasing the timing PIT towards back-loading by an amount that is small relative to windows of days or weeks.
- The one-cent tick floors longshot prices and explains much of their stickiness near the deadline.
- History before 15 August 2026 comes from series chosen by title keyword. The choice does not depend on outcomes, but it can miss one-off series with other titles.
- Strip nesting relies on the issuance order of legs.
- The CIR fit uses an average rather than a spot intensity.



(a) Event timing **(b)** Variance decomposition

Figure 13: (a) Distribution of the probability integral transform of YES resolution times under the flat hazard implied by the first priced pre-event quote; uniform (dashed) if the implied hazard shape were right. (b) Shares of the quadratic variation of the hourly mid up to settlement: diffusion (hourly changes below 20¢), jumps of at least 20¢ before resolution, and the resolution step (the YES jump plus the NO deadline step).

7 Discussion

7.1 A unified description

The three studies support one description of prediction-market prices. A binary contract's fair price is a digital option on a latent state, written either as a score, as a numeric outcome, or as the waiting time to an event. The state is revealed according to an information clock, with short-lived random variation in the rate of revelation and a large share of discrete jumps:

$$\begin{aligned} \text{single contract or ladder leg:}\quad & p_t = \mathbb{P}(Y_T > K \mid \mathcal{F}_t), \\ & \mathrm{d}Y_t = \alpha_t\, s(t)\, \mathrm{d}W_t + J_t\, \mathrm{d}N_t, \qquad \mathrm{d}\alpha_t = \nu \alpha_t\, \mathrm{d}B_t, \\ \text{deadline contract:}\quad & p_t = 1 - \mathbb{E}_t\Big[e^{-\int_t^T h_u \mathrm{d}u}\Big], \qquad h_u = h^{\mathrm{det}}(T-u)\,\xi_u, \end{aligned}$$

where h^{det} is a deterministic hazard profile that rises into the deadline and ξ a log-diffusion. Table 15 collects the estimates that pin down each component.

7.2 Why clocks differ

The information clock is the object that varies most across contract types, and it has a clear economic reading: it is the schedule on which the world reveals the answer.

- *In-game sports.* Scoring opportunities arrive roughly uniformly in game time, so a uniform clock fits, exactly as in Stern [55].
- *Pre-game sports.* Before the start almost nothing is learned, apart from occasional team news, so the clock runs at a few percent of the in-game rate.
- *Weather.* Daily temperature extremes are revealed by observations late in the local day, giving a back-loaded clock with a pronounced time-of-day profile.

Table 15: Stylized facts by model component and contract type.

Component	Estimate	Where it comes from
Variance budget	realized variance over $p_0(1-p_0)$: 0.93–1.01	weather, sports (Section 6.1); 1.19 for deadlines (Section 6.4)
Clock	uniform in-game; back-loaded weather; flat pre-game; hazard rising into deadline	Sections 6.1, 6.3 and 6.4
Backbone	$\beta \approx 0.3$–0.8; logit-normal rejected	Section 6.1
Stochastic volatility	half-life 7–21 h (weather about 40 h), $\nu \approx 0.3$–0.5 per $\sqrt{\text{day}}$	Section 6.1
Jumps	55–85% of hourly variance; about half of in-game variance at 5–15 s	Sections 6.1 and 6.2
Smile	convex, sticky in moneyness; tails too heavy	Section 6.3
Premia	implied tail mass, hazard and variance exceed realized	Sections 6.3 and 6.4

- *Deadline contracts.* The implied hazard rises into the deadline. Part of this is real, since announcements do bunch late. But realized event timing is less concentrated than prices imply, so part of the rise is a premium (Section 6.4).

For pricing, the clock is to a binary contract what the term structure of implied variance is to an option. A single contract's price carries no volatility information. A ladder's at-the-money term structure and a deadline strip's forward hazards do, and both can be compared with the clock measured from price paths.

7.3 Jumps and the limits of the Gaussian model

M0 gets the clock right and the shape wrong. Outside in-game sports, hourly standardised innovations have excess kurtosis from tens to hundreds. Even within games, roughly half of second-scale variance arrives in discrete moves. Several consequences follow.

- *Variance forecasts.* QLIKE and MSE rank the forecasting models differently: filtering the volatility state helps in quiet periods and hurts across jumps.
- *Drift.* The outward drift of the probit price per unit of total variance falls short of the Gaussian prediction, as Section 4.4 anticipates.
- *Derivative pricing.* Any instrument that depends on the path, such as the variance and first-passage claims proposed by Dalen [20], must model jumps explicitly.

The basketball result is a useful benchmark. In the NBA, jump sizes scale with the Stern factor $\varphi(Z)/\sqrt{\tau}$ almost exactly. In lower-scoring sports they scale less steeply. This is consistent with scores that are rarer, larger and more state-dependent, for which Feng et al. [29]'s Skellam model is the natural replacement.

7.4 Backbone and smile dynamics

The estimated backbone exponent of 0.3–0.8 places price volatility closer to the Wright–Fisher (Jacobi) diffusion than to the logit-normal model that is common in practice. This independently supports the Wright–Fisher component of Xi et al. [61].

In ladders, the smile moves with the implied median: volatility at a fixed strike shifts as the median moves, while at-the-money volatility does not slide along a fixed smile. In Derman [22]'s terms, the market treats the outcome distribution as a location–scale family that is re-centred on news. That is natural when news shifts the expected outcome (a score, a forecast update) without changing the remaining uncertainty much.

7.5 A tail-and-time premium

The one systematic departure from fair pricing appears in three forms:

- implied tails across strikes are heavier than realized tails;
- implied hazards exceed realized hazards at every lead before a deadline, and prices decay more slowly than the hazard model requires;
- before and during games, ladders imply more variance than is subsequently realized.

In each case the market overprices the probability of an outcome far from the centre, or of an event within a short window. This is the favourite–longshot bias of Snowberg and Wolfers [54] and Bürgi et al. [14], seen across strikes and across time.

Three features of the venue limit how far it can be corrected.

- *Tick grid.* The one-cent grid caps the gain from selling an overpriced 1–2 ¢contract at about a cent.
- *Fees.* Per-order rounding of fees makes small orders near the edges expensive.
- *Collateral.* Selling a longshot requires posting nearly the full contract value as collateral.

In line with these limits, simple rules that sell the premium earn at most a few cents per contract out of sample, and none is robust across samples. The premium is a stable feature of prices, not an exploitable inefficiency at retail scale.

Two departures from the martingale property remain.

- *Weather.* Ladder medians for daily highs drift down overnight and up during the day. This may reflect a forecast input that under-predicts daytime maxima until observations arrive.
- *Deadline contracts.* Prices fall faster than the martingale allows in their final hours, as stale prices catch up with the deadline.

Both are localised and small.

7.6 Limitations

The sample covers February to October 2026 for quotes and November 2026 for order books, a period of rapid growth on the venue. Hourly candles hide intra-hour dynamics, and the base quote panel is weighted toward actively traded markets. The order-book study covers two evenings and 50 game markets.

We do not observe the game clock or the score. In-game time to expiry is measured to scheduled settlement, which probably explains part of the cross-league dispersion. Event times of deadline contracts are upper bounds inferred from prices and closing times.

Finally, we report many cells across categories, horizons and leads. We emphasise patterns that hold in sign across the estimation and holdout samples, but individual cells should be read with multiple testing in mind.

8 Conclusion

We have proposed and calibrated an option-pricing framework for prediction-market contracts. A binary contract is a digital option on a latent state, and its price dynamics are governed by an information clock that releases a fixed variance budget $p_0(1 - p_0)$. The Gaussian version yields a Black–Scholes equation with a digital payoff and an exact transition law. A bounded SABR extension and jumps bring it to the data. Strike ladders and deadline strips extend the framework to implied distributions, smiles and hazard term structures.

On Kalshi:

- the framework describes timing well, in that clocks are stable, category-specific and forecastable;
- it describes shape poorly unless jumps are included;
- it identifies a single pricing anomaly, a premium on tail outcomes and on short time windows.

Four extensions follow naturally:

- using the game clock and score to test jump laws directly;
- longer order-book samples;
- contracts on the same outcome at two horizons, such as the first half and the full game, which would identify the clock without a model;
- pricing derivatives written on prediction-market prices, for which this framework supplies the dynamics.

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A Derivations

Proof of Lemma 1. Since p is a square-integrable martingale, $p^2 - \langle p \rangle$ is a martingale. Hence $\mathbb{E}\langle p \rangle_T = \mathbb{E}p_T^2 - p_0^2$. Because $p_T \in \{0, 1\}$, $p_T^2 = p_T$ and $\mathbb{E}p_T = p_0$, so $\mathbb{E}\langle p \rangle_T = p_0 - p_0^2$. For a discrete grid, martingale increments are orthogonal, so $\mathbb{E}\sum_i (p_{t_{i+1}} - p_{t_i})^2 = \mathbb{E}(p_T - p_0)^2 = p_0(1 - p_0)$. □

Proof of Proposition 1. *Price.* Given $\mathcal{F}_t$, $Y_T = Y_t + \int_t^T s \, \mathrm{d}W \sim \mathcal{N}(Y_t, V_t)$, so $\mathbb{P}(Y_T > 0 \mid \mathcal{F}_t) = \Phi(Y_t/\sqrt{V_t})$. Write $p(t, y) = \Phi(y/\sqrt{V_t})$. With $\dot{V}_t = -s(t)^2$ and $\zeta = y/\sqrt{V_t}$:

$$\partial_t p = \varphi(\zeta)\, y\, \tfrac{1}{2} V_t^{-3/2} s(t)^2, \qquad \partial_{yy} p = \varphi'(\zeta)/V_t = -\zeta\varphi(\zeta)/V_t.$$

Hence $\partial_t p + \frac{1}{2}s^2 \partial_{yy} p = \frac{1}{2}s^2 \varphi(\zeta)\big[yV^{-3/2} - \zeta V^{-1}\big] = 0$. The terminal condition holds because $V_t \to 0$ as $t \to T$.

Dynamics. By Itô's formula for $Z = YV^{-1/2}$,

$$\mathrm{d}Z = V^{-1/2} s \, \mathrm{d}W + Y \cdot (-\tfrac{1}{2}) V^{-3/2} \dot{V} \, \mathrm{d}t = \sqrt{\lambda} \, \mathrm{d}W + \tfrac{1}{2}\lambda Z \, \mathrm{d}t.$$

Applying Itô to $p = \Phi(Z)$:

$$\mathrm{d}p = \varphi(Z)\big(\sqrt{\lambda} \, \mathrm{d}W + \tfrac{1}{2}\lambda Z \, \mathrm{d}t\big) + \tfrac{1}{2}\varphi'(Z)\, \lambda \, \mathrm{d}t = \varphi(Z)\sqrt{\lambda} \, \mathrm{d}W,$$

since $\varphi'(z) = -z\varphi(z)$.

Transition. $Y_{t+h} = Y_t + \xi$ with $\xi \sim \mathcal{N}(0, V_t - V_{t+h})$ independent of $\mathcal{F}_t$. Then

$$Z_{t+h} = \frac{Y_t + \xi}{\sqrt{V_{t+h}}} = Z_t \sqrt{R} + \frac{\xi}{\sqrt{V_{t+h}}},$$

and $\mathrm{Var}\,(\xi/\sqrt{V_{t+h}}) = (V_t - V_{t+h})/V_{t+h} = R - 1$. The distribution of p_{t+h} follows because Φ is increasing. □

Uniform clock. With s constant, $V_t = s^2(T - t)$, $\lambda_t = 1/(T - t)$ and $R = \tau/(\tau - h)$. For a power-law clock $V_t \propto \tau^\gamma$, $\lambda_t = \gamma/\tau$. A ladder whose at-the-money implied volatility scales as $\sigma_{\mathrm{ATM}} \propto \tau^b$ therefore implies $\lambda_t = 2b/\tau$; Brownian information gives $b = \frac{1}{2}$.

Backbone asymptotics. Near zero, Mills' ratio gives $\varphi(\Phi^{-1}(p)) = p\,|\Phi^{-1}(p)|\,(1 + o(1))$, and $|\Phi^{-1}(p)| = \sqrt{2\ln(1/p)}\,(1 + o(1))$. So $\varphi(\Phi^{-1}(p)) \sim p\sqrt{2\ln(1/p)}$, which lies between $g_1(p) \sim p$ and $g_{1/2}(p) \sim \sqrt{p}$.

Proof of Proposition 2. (i) If $X_T \mid \mathcal{F}_t \sim \mathcal{N}(F, \sigma^2)$ then $S(K) = \Phi((F-K)/\sigma)$ and $\sigma_K = \sigma$ for every K.

(ii) Let $X_T = F + \sigma_0 \varepsilon$ with ε symmetric, with distribution G and upper quartile x_q, so that $G(x_q) = \frac{3}{4}$. Take $K > F$ and $x = (K-F)/\sigma_0$. Then $S(K) = 1 - G(x)$ and $\sigma_K = \sigma_0\, x/\Phi^{-1}(G(x))$. Matching the interquartile range defines $\sigma_{\mathrm{ATM}} = \sigma_0 x_q/\Phi^{-1}(\frac{3}{4})$, so

$$\frac{\sigma_K}{\sigma_{\mathrm{ATM}}} = \frac{x}{x_q} \cdot \frac{\Phi^{-1}(\frac{3}{4})}{\Phi^{-1}(G(x))}.$$

Heavier tails than the quartile-matched Gaussian means $1 - G(x) > 1 - \Phi\big(x\,\Phi^{-1}(\frac{3}{4})/x_q\big)$ for $x > x_q$. That is, $\Phi^{-1}(G(x)) < x\,\Phi^{-1}(\frac{3}{4})/x_q$, which gives $\sigma_K/\sigma_{\mathrm{ATM}} > 1$. The case $K < F$ follows by symmetry. For a Student-t with ν degrees of freedom, $x/\Phi^{-1}(G(x))$ is increasing in $x > 0$. We verified this numerically on a fine grid for $\nu \in [1.5, 100]$; we do not prove it in general. This gives monotonicity in $|m|$.

(iii) Under right skew the upper tail is heavier than the lower one, so by the same argument applied to each tail separately, σ_K is larger for $m > 0$ than for $m < 0$ at equal $|m|$. □

Proof of (9). If $X_T \mid \mathcal{F}_t \sim \mathcal{N}(F_t, \sigma_{\mathrm{ATM}}(t)^2)$ for all t, then $F_t = \mathbb{E}[X_T \mid \mathcal{F}_t]$ is a martingale. By the conditional-variance decomposition,

$$\sigma_{\mathrm{ATM}}(t)^2 = \mathbb{E}_t\big[\sigma_{\mathrm{ATM}}(t+h)^2\big] + \mathrm{Var}_t(F_{t+h}).$$

Taking expectations gives (9). Without Gaussianity the identity holds for the conditional mean and variance. The empirical version uses the median and the interquartile scale as robust proxies. □

Proof of Proposition 3. On $\{\vartheta > t\}$, $\mathbb{P}(\vartheta \le T \mid \mathcal{F}_t) = 1 - \mathbb{E}_t \exp(-\int_t^T h)$ is the standard reduced-form survival formula [43]. For deterministic h, $p_t = 1 - \exp(-H(t,T))$ with $\partial_t H = -h_t$, so $\dot{p}_t = -h_t e^{-H} = -h_t(1-p_t)$.

Write $D_t = \mathbf{1}\{\vartheta \le t\}$. The price process is $p_t = D_t + (1-D_t)(1 - e^{-H(t,T)})$, and

$$\mathrm{d}p_t = (1-D_{t-})\big[-h_t(1-p_{t-})\,\mathrm{d}t + (1-p_{t-})\,\mathrm{d}D_t\big] = (1-D_{t-})(1-p_{t-})\big(\mathrm{d}D_t - h_t\,\mathrm{d}t\big).$$

This is a martingale because $D_t - \int_0^{t\wedge\vartheta} h_u\,\mathrm{d}u$ is one. For the term structure, $1 - p_t(T_i) = \exp(-H(t,T_i))$, so $H(T_1,T_2) = H(t,T_2) - H(t,T_1) \ge 0$, and the forward hazard formula follows. □

B Additional results: single-contract dynamics

Variance identity. Table 16 reports the market-level check of the variance identity behind Section 6.1.1. The path share falls as the sampling interval grows, because two-sided quotes with spreads of at most 4 ¢ become sparse near resolution. For sports the identity holds even though 93–95% of the variance is realized after the last such quote, mostly in-game and at the final whistle. Sampling every hour or every 24 hours gives the same picture. For weather the total ratio is 0.96, 0.97 and 0.96 on the estimation sample at 1, 6 and 24 hours.

Backbone fits. Table 17 gives the full backbone comparison behind Section 6.1.4 and Figure 3.

Table 16: Variance identity by category

Category	Split	Markets	Events	$\mathbb{E}(y-p_0)^2/\mathbb{E}\,p_0(1-p_0)$ (t)	Total ratio (t)	Path share	Terminal share
Sports (all)	Est.	9,533	8,551	0.99 (−1.8)	0.98 (−2.0)	0.07	0.93
Sports (all)	Holdout	10,007	8,991	1.01 (1.5)	1.01 (1.3)	0.05	0.95
Weather	Est.	5,838	1,183	0.96 (−3.0)	0.97 (−1.7)	0.29	0.70
Weather	Holdout	4,595	938	0.94 (−3.2)	0.93 (−3.1)	0.30	0.68
Mentions	Est.	336	109	1.05 (0.8)	1.00 (0.0)	0.27	0.73
Mentions	Holdout	478	98	1.20 (4.0)	1.30 (4.7)	0.23	0.82
Politics	Est.	274	102	1.34 (2.7)	1.21 (1.9)	0.68	0.44
Politics	Holdout	534	167	1.15 (1.5)	1.26 (2.5)	0.77	0.39
Elections	Est.	363	100	0.94 (−0.8)	0.86 (−1.6)	0.37	0.57
Elections	Holdout	369	120	1.30 (2.7)	1.16 (1.6)	0.72	0.38
Entertainment	Est.	438	174	1.24 (2.7)	1.37 (3.5)	0.84	0.39
Entertainment	Holdout	375	129	1.08 (0.8)	1.06 (0.6)	0.80	0.24
Sci. & Tech.	Est.	333	96	1.19 (1.5)	1.25 (1.9)	0.85	0.32
Sci. & Tech.	Holdout	382	129	1.53 (3.7)	1.62 (3.9)	1.25	0.23

One row per market; quotes with spread $\leq 4\,¢$ sampled every 6 hours. p_0 is the first such mid and y the settled outcome. *Total ratio*: $\mathbb{E}[\sum(\Delta p)^2 + (y - p_{\text{last}})^2]/\mathbb{E}[p_0(1-p_0)]$, which equals one for a calibrated martingale (Section 4). *Path* and *terminal share*: the two parts of the numerator relative to it.
Split by close date: estimation sample (before 1 October 2026) and holdout (October 2026). This check fits no parameters, so Elections has a holdout here. t for the mean difference from $p_0(1-p_0)$, clustered by event.
Candle samples outside sports and weather were partly selected on trading volume (ex post), which biases their ratios upward.

C Additional results: strike ladders

This appendix reports the ladder sample by group (Table 18), the smile parameters for every group, lead and sample (Table 19), the ATM term-structure exponents (Table 20), the tail-mass comparison underlying Figure 9 (Table 21), the variance ratios underlying Figure 10 (Table 22) and the economic-significance calculation (Table 23).

D Additional results: deadline contracts

Calibration by category and for longshots. Table 24 breaks the calibration of Table 12 down by category, and Table 25 reports the two cheapest price buckets by lead. Below 5 ¢ the hazard multiplier is bounded below at 0.02 by construction when a cell has no YES outcomes, and the one-cent tick bounds the mid from below, so those cells mainly describe the price floor.

Event timing by type, category and starting price. Table 26 extends the timing evidence of Section 6.4.4. Relative to the flat implied hazard, the mean PIT exceeds 0.5 in every category with at least 15 resolutions and in every starting-price bucket, and significantly so everywhere except elections ($t = 1.79$). Sports and entertainment, where resolutions often hinge on scheduled games or releases, are the most back-loaded.

Variance decomposition and Cox intensity. Table 27 gives the quadratic-variation shares behind Figure 13b, and Table 28 the CIR fit of Equation (25). The negative mean-reversion estimates at both horizons are the formal counterpart of the rising single-contract hazard paths in Figure 12a.

Economic significance. Table 29 reports the trading test summarised in Section 6.4.7. Thresholds are chosen on the estimation sample by the highest mean profit with at least 30 trades, so the estimation sample columns are in-sample.

Table 17: Backbone estimates and QLIKE comparisons

	β			Est., vs. probit (t)		Holdout, vs. probit (t)		
Category	QLIKE	Roll (s.e.)	Roll, hold.	Logit	Power	Logit	Power	Jacobi
Sports in-game	0.47	0.49 (0.14)	0.35	0.016 (4.9)	−0.020 (−1.8)	0.032 (4.7)	−0.058 (−2.5)	−0.054 (−2.7)
Sports pre-game	0.39	0.44 (0.04)	0.26	0.044 (6.2)	−0.044 (−3.5)	0.070 (1.6)	−0.090 (−1.6)	−0.075 (−1.7)
Weather	0.58	0.54 (0.04)	0.69	0.054 (5.5)	−0.029 (−2.7)	0.028 (2.8)	−0.018 (−1.8)	−0.009 (−0.7)
Mentions	0.63	0.76 (0.11)	0.60	0.045 (3.5)	−0.014 (−1.1)	0.247 (1.2)	−0.187 (−1.0)	−0.257 (−0.9)
Politics	0.39	0.49 (0.14)	0.82	0.075 (4.0)	−0.078 (−1.5)	0.161 (2.4)	−0.144 (−1.2)	−0.142 (−1.4)
Elections	0.81	0.86 (0.10)	–	0.028 (1.0)	0.000 (−0.1)	–	–	–
Entertainment	0.65	0.62 (0.05)	0.95	0.040 (5.2)	−0.014 (−1.9)	−0.005 (−0.2)	0.041 (1.6)	0.108 (2.0)
Sci. & Tech.	0.31	0.51 (0.12)	0.93	0.137 (7.8)	−0.203 (−2.9)	0.081 (1.0)	−0.572 (−1.2)	−0.595 (−1.3)

Hourly steps; estimation sample (before 1 October 2026) and holdout (October 2026). *QLIKE*: β of the power backbone in Equation (13), fitted on the estimation sample. *Roll*: log cell-mean regression on Roll-corrected $(\Delta p)^2$, with τ-bucket fixed effects; standard errors from an event-level Poisson bootstrap (100 draws).
QLIKE differences are mean per-step losses relative to the probit backbone of M0 (negative: better than probit). Logit is $\beta = 1$, Jacobi is $\beta = 1/2$, and "Power" uses the estimated β. t clustered by event.

Table 18: Ladder sample by group.

Group	Series	Events	Ladders	Snapshots	Leg quotes	Estimation	Holdout
Weather highs	24	1,531	1,531	5,762	34,572	3,286	2,476
Weather lows	24	836	836	1,509	9,054	801	708
Sports totals	149	3,778	3,778	7,762	78,874	4,245	3,517
Team totals	24	1,040	2,004	4,570	44,560	2,279	2,291
Spreads	44	1,760	1,760	4,010	61,022	1,740	2,270
Politics	11	115	115	270	2,157	110	160
Entertainment	11	82	82	207	1,775	122	85
Science/tech	14	33	33	57	280	34	23
Other climate	7	37	37	44	297	10	34
Elections	1	1	1	1	3	0	1

Fitted snapshots (at least three survival points, median inside the quoted strikes). Leg quotes: legs with spread at most 5 ¢ used in the fits. Estimation and holdout columns count snapshots of events expiring before 1 October 2026 and in October 2026. Team-total events carry one ladder per team.

Table 19: Smile parameters by group, lead and sample.

Group	Lead	Snapshots	Strikes	s [95% CI]	c [95% CI]	Median s	Median c
Weather highs	36 h, est.	668	1,758	−0.017 [−0.020, −0.013]	0.053 [0.051, 0.056]	−0.024	0.033
Weather highs	36 h, hold.	489	1,162	−0.018 [−0.021, −0.014]	0.058 [0.056, 0.060]	−0.025	0.033
Weather highs	24 h, est.	734	1,668	−0.019 [−0.022, −0.015]	0.054 [0.052, 0.056]	−0.032	0.035
Weather highs	24 h, hold.	538	1,042	−0.015 [−0.018, −0.011]	0.056 [0.054, 0.059]	−0.023	0.033
Weather highs	16 h, est.	741	1,348	−0.019 [−0.023, −0.016]	0.054 [0.051, 0.056]	−0.036	0.035
Weather highs	16 h, hold.	553	876	−0.011 [−0.015, −0.008]	0.055 [0.052, 0.057]	−0.019	0.030
Weather highs	12 h, est.	520	603	0.002 [−0.003, 0.006]	0.058 [0.054, 0.062]	−0.022	0.030
Weather highs	12 h, hold.	357	302	−0.004 [−0.011, 0.003]	0.053 [0.048, 0.058]	0.004	0.033
Weather lows	36 h, est.	87	205	−0.030 [−0.043, −0.018]	0.069 [0.061, 0.077]	−0.041	0.038
Weather lows	36 h, hold.	15	33	−0.032 [−0.061, −0.003]	0.058 [0.046, 0.070]	−0.014	0.037
Sports totals	48 h, est.	123	1,160	0.044 [0.037, 0.051]	0.025 [0.018, 0.032]	0.054	0.004
Sports totals	48 h, hold.	175	1,498	0.057 [0.047, 0.068]	0.038 [0.015, 0.061]	0.059	0.005
Sports totals	24 h, est.	986	7,702	0.068 [0.065, 0.072]	0.021 [0.017, 0.024]	0.086	0.001
Sports totals	24 h, hold.	1,253	9,841	0.058 [0.056, 0.061]	0.022 [0.018, 0.026]	0.073	0.000
Sports totals	6 h, est.	2,141	13,196	0.088 [0.085, 0.092]	0.007 [0.005, 0.010]	0.098	−0.007
Sports totals	6 h, hold.	1,574	12,494	0.057 [0.055, 0.059]	0.018 [0.014, 0.021]	0.071	−0.004
Sports totals	1 h, est.	995	4,954	0.117 [0.107, 0.127]	0.026 [0.019, 0.033]	0.155	−0.024
Sports totals	1 h, hold.	515	4,277	0.050 [0.044, 0.056]	0.039 [0.036, 0.042]	0.072	0.019
Team totals	48 h, est.	103	806	0.000 [−0.007, 0.007]	0.037 [0.033, 0.042]	0.004	0.024
Team totals	48 h, hold.	173	1,320	0.008 [0.004, 0.012]	0.036 [0.033, 0.039]	0.003	0.019
Team totals	24 h, est.	730	4,246	0.052 [0.044, 0.059]	0.043 [0.032, 0.053]	0.075	−0.002
Team totals	24 h, hold.	851	6,758	0.035 [0.032, 0.039]	0.022 [0.018, 0.025]	0.038	0.002
Team totals	6 h, est.	1,023	6,589	0.061 [0.056, 0.067]	0.028 [0.024, 0.031]	0.119	−0.011
Team totals	6 h, hold.	958	9,148	0.035 [0.031, 0.038]	0.020 [0.016, 0.024]	0.036	−0.006
Team totals	1 h, est.	423	1,555	0.114 [0.098, 0.131]	0.041 [0.035, 0.048]	0.202	−0.017
Team totals	1 h, hold.	309	2,247	0.028 [0.020, 0.035]	0.050 [0.046, 0.054]	0.038	0.026
Spreads	48 h, est.	95	1,491	0.030 [0.014, 0.046]	0.056 [0.049, 0.062]	−0.001	0.031
Spreads	48 h, hold.	151	1,643	0.017 [−0.004, 0.037]	0.055 [0.025, 0.085]	−0.012	0.025
Spreads	24 h, est.	501	6,486	0.019 [0.013, 0.025]	0.045 [0.040, 0.050]	0.002	0.017
Spreads	24 h, hold.	735	10,167	0.009 [0.003, 0.016]	0.031 [0.023, 0.038]	−0.002	0.012
Spreads	6 h, est.	755	7,711	0.017 [0.011, 0.023]	0.053 [0.038, 0.068]	−0.005	0.008
Spreads	6 h, hold.	966	12,318	0.006 [0.001, 0.012]	0.036 [0.025, 0.047]	−0.002	0.010
Spreads	1 h, est.	389	2,638	0.010 [−0.001, 0.021]	0.062 [0.055, 0.068]	−0.005	0.030
Spreads	1 h, hold.	418	4,467	−0.011 [−0.017, −0.006]	0.037 [0.034, 0.039]	−0.016	0.023
Politics	48 h, est.	25	69	0.034 [−0.002, 0.070]	0.057 [0.045, 0.069]	0.078	0.044
Politics	48 h, hold.	28	108	0.033 [0.011, 0.055]	0.065 [0.052, 0.079]	0.014	0.043
Politics	24 h, est.	35	105	−0.001 [−0.022, 0.021]	0.083 [0.071, 0.095]	−0.012	0.071
Politics	24 h, hold.	51	224	0.015 [−0.004, 0.034]	0.068 [0.055, 0.081]	0.007	0.052
Politics	6 h, est.	32	76	0.002 [−0.025, 0.028]	0.066 [0.055, 0.077]	0.001	0.072
Politics	6 h, hold.	51	177	0.010 [−0.013, 0.032]	0.074 [0.064, 0.084]	0.013	0.060
Science/tech	48 h, hold.	14	34	0.012 [−0.030, 0.054]	0.061 [0.037, 0.086]	0.126	0.125
Other climate	24 h, hold.	26	126	0.156 [0.135, 0.178]	0.019 [0.003, 0.035]	0.164	0.009

Pooled least squares of $\sigma_K/\sigma_{\text{ATM}} - 1 = a + s\,m + c\,m^2$ over off-ATM strikes ($|z| \geq 0.25$, $|m| \leq 3.5$, the two strikes straddling F excluded) in cells with at least 30 strikes; est. = estimation sample (before 1 October 2026), hold. = holdout (October 2026); 95% intervals from standard errors clustered by event. Median s, c: medians of the per-snapshot fits with $a = 0$ (snapshots with at least three off-ATM strikes).

Table 20: ATM term structure: exponent of σ_{ATM} in the lead.

Group	Snapshots	Ladders	η	95% CI	$t_{\eta=1/2}$
Weather highs	5,640	1,409	0.773	[0.760, 0.786]	41.3
Weather lows	1,162	489	0.480	[0.413, 0.546]	−0.6
Sports totals	4,749	2,227	0.009	[0.007, 0.010]	−595.8
Team totals	3,430	1,580	0.001	[−0.003, 0.005]	−261.1
Spreads	2,706	1,231	0.001	[−0.002, 0.004]	−325.9
Politics	245	90	0.348	[0.288, 0.408]	−5.0
Entertainment	184	59	0.186	[0.138, 0.234]	−12.7
Science/tech	38	14	0.241	[0.170, 0.311]	−7.2

Within-ladder regression of $\log \sigma_{\mathrm{ATM}}$ on the log lead (ladders with at least two snapshots), estimation sample and holdout pooled; intervals and t clustered by event. Sports use the pre-game leads (48, 24, 6 hours); weather leads are hours before the measurement-day close.

Table 21: Implied and realised tail mass beyond $|m|$ (cents per strike).

		Lower wing ($m \le -k$)					Upper wing ($m \ge k$)				
Group	Sample, k	Strikes	Implied	Realised	Normal	z_{H_0}	Strikes	Implied	Realised	Normal	z_{H_0}
Weather highs	est., 1.5	5,031	2.17	0.97	1.07	0.83	5,545	1.85	0.63	1.12	−1.28
Weather highs	est., 2.5	2,953	1.11	0.30	0.09	−0.50	3,386	0.95	0.03	0.09	−2.57
Weather highs	hold., 1.5	4,066	1.86	0.81	0.95	0.35	4,539	1.61	0.48	0.95	−1.74
Weather highs	hold., 2.5	2,580	1.05	0.00	0.09	−2.37	2,959	0.93	0.00	0.07	−2.29
Weather lows	est., 1.5	2,221	2.14	0.54	0.95	−2.23	1,425	0.96	0.00	0.58	−1.98
Weather lows	est., 2.5	1,552	1.54	0.32	0.01	−2.05	949	0.65	0.00	0.14	−1.54
Weather lows	hold., 1.5	2,083	1.64	0.48	0.82	−1.49	1,338	0.87	0.07	0.51	−1.49
Weather lows	hold., 2.5	1,471	1.14	0.14	0.01	−1.54	925	0.66	0.00	0.15	−1.45
Sports totals	est., 1.5	936	3.45	2.35	4.05	−0.96	2,200	5.90	6.18	3.35	0.67
Sports totals	est., 2.5	157	0.66	0.64	0.24	−0.06	456	1.77	2.19	0.18	0.45
Sports totals	hold., 1.5	3,018	2.21	1.29	2.69	−0.71	3,476	5.08	3.22	3.02	−0.77
Sports totals	hold., 2.5	858	0.55	0.00	0.12	−1.20	914	1.70	0.11	0.15	−2.17
Team totals	est., 1.5	1,086	3.19	2.76	3.33	−0.30	1,930	5.95	5.85	3.09	0.44
Team totals	est., 2.5	158	0.80	0.00	0.20	−0.83	362	3.62	1.10	0.19	−2.03
Team totals	hold., 1.5	1,933	3.35	2.85	3.24	0.34	2,551	4.51	4.74	2.96	1.46
Team totals	hold., 2.5	334	1.05	0.00	0.18	−1.00	568	2.06	0.70	0.17	−0.89
Spreads	est., 1.5	2,097	4.46	6.82	3.17	0.59	501	5.88	5.79	3.72	0.65
Spreads	est., 2.5	539	1.60	2.78	0.14	1.10	90	1.26	1.11	0.17	0.39
Spreads	hold., 1.5	4,000	3.65	2.83	2.86	−1.02	1,425	4.30	4.35	3.44	−0.63
Spreads	hold., 2.5	1,135	1.04	0.26	0.14	−1.74	193	1.63	0.00	0.24	−1.23
Politics	est., 1.5	197	3.09	2.03	1.26	0.10	320	2.96	2.50	0.87	1.29
Politics	est., 2.5	123	1.86	0.00	0.07	−0.84	232	2.05	1.29	0.04	0.22
Politics	hold., 1.5	343	3.23	0.87	1.14	−0.32	454	3.32	1.54	0.78	−0.06
Politics	hold., 2.5	222	2.21	0.90	0.07	−0.37	338	2.38	0.00	0.05	−1.23
Entertainment	est., 1.5	343	1.14	0.87	0.57	−0.30	478	1.20	0.21	0.47	−0.80
Entertainment	est., 2.5	280	0.76	0.71	0.08	−0.38	402	0.94	0.00	0.05	−0.68
Entertainment	hold., 1.5	383	1.02	0.26	0.39	1.01	485	1.03	0.00	0.30	−0.68
Entertainment	hold., 2.5	331	0.76	0.00	0.04	−0.53	433	0.84	0.00	0.03	−0.56

All strikes beyond the threshold, all leads; est. = estimation sample (before 1 October 2026), hold. = holdout (October 2026). Implied: $S(K)$ in the upper wing, $1 - S(K)$ in the lower wing; realised: the corresponding settled indicator; Normal: $\Phi(-|m|)$, the tail of a normal with the ATM width. z_{H_0}: event means of realised minus implied over the variance implied by the quotes (events independent), which remains valid when no tail event occurs.

Table 22: Realised and implied variance between consecutive snapshots.

Group	Window	Pairs	Events	RV	IV	RV/IV	90% CI	Drift (t)
Weather highs	36→24	1,034	1,034	0.1356	0.1324	1.024	[0.867, 1.243]	0.065 (5.8)
Weather highs	24→16	1,142	1,142	0.1259	0.1922	0.655	[0.584, 0.731]	−0.046 (−4.5)
Weather highs	16→12	816	816	0.3329	0.3070	1.084	[0.946, 1.221]	0.060 (3.0)
Weather highs	16→8	339	339	0.8501	0.8345	1.019	[0.901, 1.153]	0.176 (3.6)
Weather highs	12→8	713	713	1.0595	0.6849	1.547	[1.374, 1.738]	0.106 (2.8)
Weather lows	16→12	236	236	0.0206	0.0709	0.291	[0.248, 0.332]	0.084 (11.0)
Weather lows	12→8	320	320	0.0208	0.0251	0.828	[0.259, 1.890]	0.036 (4.6)
Sports totals	48→24	296	296	0.0023	0.0053	–	–	0.003 (0.9)
Sports totals	24→6	2,225	2,225	0.0041	0.0177	0.233	[0.189, 0.290]	0.003 (2.0)
Sports totals	6→1	1,462	1,462	0.2797	0.3122	0.896	[0.810, 0.982]	0.022 (1.6)
Team totals	48→24	270	142	0.0012	−0.0202	–	–	−0.002 (−0.8)
Team totals	24→6	1,575	822	0.0033	−0.0329	–	–	−0.001 (−0.6)
Team totals	6→1	716	434	0.3507	0.5417	0.647	[0.590, 0.701]	−0.009 (−0.4)
Spreads	48→24	244	244	0.0019	−0.0326	–	–	0.003 (1.1)
Spreads	24→6	1,230	1,230	0.0040	−0.0088	–	–	0.001 (0.5)
Spreads	6→1	773	773	0.4392	0.4786	0.918	[0.835, 1.040]	0.013 (0.6)
Politics	48→24	46	46	0.3432	0.0819	4.191	[0.760, 23.987]	−0.099 (−1.1)
Politics	24→6	68	68	0.2657	0.2554	1.040	[0.367, 3.623]	−0.027 (−0.4)
Politics	6→1	32	32	1.5147	0.6672	2.270	[1.187, 4.407]	0.576 (3.0)

Consecutive snapshots $t \to u$ of one ladder (window in hours before expiry; weather windows with at least 100 pairs), estimation sample and holdout pooled. RV: mean of $((F_u - F_t)/\sigma_t)^2$; IV: mean of $1 - (\sigma_u/\sigma_t)^2$, with $\sigma = \sigma_{\text{ATM}}$. RV/IV is reported where IV exceeds 0.01; 90% interval from 200 cluster-bootstrap replications over events. Drift: mean of $(F_u - F_t)/\sigma_t$ with t clustered by event. Sports windows from 6 to 1 hours are in play.

Table 23: Economic significance: selling the tail side of ladder legs.

		Estimation sample			Holdout		
Rule	Tail price	Contracts	PnL	t_{risk}	Contracts	PnL	t_{risk}
Every wing leg	23.67	92,217	−2.30	−2.90	108,284	−1.75	−2.11
Model, $\lvert m\rvert \geq 1.5$	4.39	12,977	−0.78	−1.63	17,443	−0.22	−0.46
Model, $\lvert m\rvert \geq 2$	2.76	5,227	0.02	0.04	6,491	0.65	1.39
Model, $\lvert m\rvert \geq 2.5$	2.36	2,741	0.23	0.43	3,006	1.10	1.99
Price, tail $\leq 2\,¢$	1.61	2,572	0.31	0.73	3,821	0.25	0.65
Price, tail $\leq 3\,¢$	2.08	4,650	0.22	0.53	6,924	−0.04	−0.09
Price, tail $\leq 5\,¢$	3.08	9,510	−0.47	−1.12	12,913	−0.21	−0.49
Price, tail $\leq 10\,¢$	5.49	20,252	−1.26	−2.56	27,288	−0.91	−1.81
Price $\leq 5\,¢$ and $\lvert m\rvert < 2$	3.71	4,800	−1.22	−2.37	6,757	−0.85	−1.65
Price $\leq 5\,¢$ and $\lvert m\rvert \geq 2$	2.41	4,710	0.29	0.62	6,156	0.49	1.08

Pre-specified rules pooled over all groups and leads, not tuned. Each qualifying leg at each snapshot is one contract: the side opposite the tail is bought at the snapshot candle's ask (or 1−bid) with the Kalshi taker fee of its series and held to settlement. Tail price: mean mid of the tail side (cents). PnL in cents per contract. t_{risk}: mean over its standard error under settlement risk at the market price (per-contract s.d. $\sqrt{p(1-p)}$ summed within an event, events independent). Estimation sample: before 1 October 2026; holdout: October 2026.

Table 24: Calibration of implied hazards by category (events, mentions and counts)

Category	Lead	n (cl.)	Mid	k [90% int.]	$\bar{y} - \bar{p}$ (¢; t)
Politics	$<$ 1 d	2,387 (274)	0.108	0.50 [0.40, 0.62]	−2.18 (−5.61)
	1–7 d	2,076 (206)	0.200	0.64 [0.52, 0.84]	−3.98 (−3.10)
	$\geq$ 7 d	2,021 (210)	0.228	0.65 [0.54, 0.79]	−5.39 (−4.33)
Mentions	$<$ 1 d	3,822 (70)	0.236	0.65 [0.58, 0.74]	−1.74 (−5.39)
	1–7 d	4,181 (66)	0.353	0.86 [0.77, 0.95]	−1.75 (−2.47)
	$\geq$ 7 d	1,048 (21)	0.411	0.89 [0.76, 1.05]	−2.00 (−1.25)
Science & tech.	$<$ 1 d	687 (51)	0.032	0.23 [0.12, 0.36]	−2.16 (−6.14)
	1–7 d	399 (41)	0.138	0.55 [0.32, 0.82]	−4.49 (−2.19)
	$\geq$ 7 d	587 (47)	0.233	0.59 [0.43, 0.84]	−6.39 (−2.69)
Sports	$<$ 1 d	404 (37)	0.112	0.58 [0.45, 0.72]	−1.58 (−3.49)
	1–7 d	266 (31)	0.271	0.82 [0.67, 1.25]	−2.33 (−1.11)
	$\geq$ 7 d	242 (29)	0.323	1.02 [0.76, 1.79]	0.33 (0.10)
Entertainment	$<$ 1 d	429 (66)	0.058	0.85 [0.35, 1.56]	−0.39 (−0.37)
	1–7 d	273 (41)	0.214	0.64 [0.44, 1.06]	−3.12 (−1.65)
	$\geq$ 7 d	411 (52)	0.295	0.38 [0.28, 0.57]	−12.95 (−4.08)
Elections	$<$ 1 d	327 (31)	0.050	0.97 [0.47, 1.80]	−0.08 (−0.07)
	1–7 d	190 (24)	0.150	1.15 [0.60, 2.27]	1.35 (0.36)
	$\geq$ 7 d	229 (28)	0.217	0.81 [0.43, 1.56]	−2.93 (−0.59)
Climate & weather	$<$ 1 d	37 (6)	0.140	0.85 [0.43, 1.56]	−0.47 (−0.42)
	1–7 d	52 (8)	0.170	2.00 [0.35, 10.45]	4.17 (0.51)
	$\geq$ 7 d	60 (10)	0.231	0.91 [0.36, 2.36]	−1.44 (−0.16)

As Table 12, all splits pooled, eliminations excluded. Four categories with fewer than four contracts (culture, health, social, world) are omitted.

Table 25: Calibration of cheap contracts by lead (events, mentions and counts)

	Mid below 5¢				Mid 5–15¢			
Lead	n (cl.)	Mid	YES	k [90% int.]	n (cl.)	Mid	YES	k [90% int.]
$<$ 6 h	3,388 (482)	0.013	0.001	0.04 [0.02, 0.13]	315 (108)	0.078	0.051	0.64 [0.35, 1.01]
6–24 h	2,456 (353)	0.018	0.002	0.11 [0.02, 0.20]	600 (138)	0.082	0.045	0.54 [0.33, 0.77]
1–3 d	1,565 (303)	0.023	0.010	0.45 [0.27, 0.69]	894 (179)	0.084	0.049	0.58 [0.43, 0.74]
3–7 d	952 (269)	0.023	0.011	0.45 [0.22, 0.71]	869 (216)	0.089	0.074	0.82 [0.61, 1.07]
7–14 d	571 (174)	0.025	0.011	0.42 [0.21, 0.78]	501 (163)	0.086	0.052	0.59 [0.36, 0.84]
14–30 d	314 (123)	0.025	0.016	0.64 [0.19, 1.17]	393 (160)	0.089	0.041	0.44 [0.27, 0.64]
30–60 d	230 (73)	0.028	0.000	$\leq$ 0.02	246 (101)	0.093	0.033	0.34 [0.14, 0.58]

As Table 12, all splits pooled. YES is the share of observations whose contract resolved YES. k is reported as $\leq$ 0.02 when a cell has no YES outcome.

Table 26: Timing of YES resolutions relative to the flat implied hazard

	n (cl.)	p_0	Anchor (h)	Mean PIT (t)	KS	Last 10%: real./flat	Last day
Events, mentions, counts	807 (238)	0.561	120	0.61 (7.72)	0.16	0.157 / 0.064	0.201
One-off events	304 (151)	0.560	504	0.65 (8.34)	0.22	0.197 / 0.063	0.207
Mentions	447 (66)	0.566	120	0.60 (4.56)	0.16	0.132 / 0.064	0.201
Counts	56 (21)	0.525	120	0.52 (0.35)	0.13	0.143 / 0.066	0.161
Eliminations	92 (45)	0.510	120	0.95 (19.03)	0.81	0.837 / 0.067	0.913
Politics	223 (102)	0.559	504	0.60 (4.05)	0.14	0.152 / 0.063	0.157
Science & tech.	44 (18)	0.609	420	0.64 (2.46)	0.33	0.114 / 0.060	0.136
Sports	39 (15)	0.517	504	0.74 (6.59)	0.38	0.359 / 0.067	0.385
Elections	32 (18)	0.497	204	0.60 (1.79)	0.23	0.219 / 0.067	0.281
Entertainment	15 (14)	0.583	1080	0.78 (6.52)	0.44	0.267 / 0.063	0.267
$p_0 < 0.15$	42 (33)	0.089	120	0.64 (2.66)	0.24	0.238 / 0.096	0.238
$0.15 \leq p_0 < 0.5$	246 (132)	0.327	120	0.60 (4.57)	0.16	0.175 / 0.083	0.228
$0.5 \leq p_0 < 0.9$	519 (191)	0.710	120	0.62 (7.07)	0.17	0.143 / 0.052	0.185

Contracts that resolved YES, anchored at their earliest priced pre-event grid quote with mid below 0.9 (p_0: mean anchor price; Anchor: median anchor lead). PIT: $F(u) = [1 - (1 - p_0)^u]/p_0$ at the elapsed window share u of the event-time upper bound; uniform under the flat implied hazard. t tests mean PIT = 0.5 with clustering by event. KS: Kolmogorov–Smirnov distance from uniform. Last 10%: share of resolutions in the last tenth of the window, realised and under the flat implied hazard. Last day: share within 24 hours of the deadline. Category and price rows cover events, mentions and counts. All splits pooled.

Table 27: Quadratic variation of the hourly mid up to settlement

				Share of quadratic variation					
Type	Contracts	YES	Hours	Diff.	Jumps	YES res.	NO step	QV ratio	YES jump
Events, mentions, counts	4,032	0.405	133	0.344	0.423	0.225	0.008	1.19	0.162
One-off events	1,983	0.267	181	0.372	0.454	0.163	0.011	1.09	0.156
Mentions	1,778	0.564	113	0.352	0.397	0.250	0.001	1.23	0.148
Counts	271	0.373	117	0.160	0.433	0.374	0.033	1.50	0.326
Eliminations	1,388	0.105	13	0.256	0.689	0.045	0.010	0.96	0.056

Hourly mids with spread at most 10¢ from the first quote in the last 62 days before the deadline; contracts with at least six such hours. YES: share resolving YES. Hours: median count. Diff.: hourly changes below 20¢ before resolution. Jumps: hourly changes of at least 20¢ before resolution. YES res.: $(1 - m)^2$ from the last mid at least two hours before the event-time upper bound. NO step: m^2 from the last mid before the deadline. QV ratio: $\sum QV / \sum p_0(1 - p_0)$, which is one in expectation for a martingale. YES jump: mean $1 - m$ for contracts that resolved YES. All splits pooled.

Table 28: CIR-type fit of the implied average hazard on no-event steps

Type	Leads	Steps	Contracts	Median $\bar{h}$ (/d)	κ (/d)	θ (/d)	σ	$\sigma_{\ln \bar{h}}$
Events, mentions, counts	≥ 3 d	5,034	1,987	0.023	−0.049	−0.0002	0.23	0.42
	< 3 d	3,729	1,279	0.113	−1.987	0.0146	8.16	1.45
One-off events	≥ 3 d	2,660	924	0.014	−0.060	0.0003	0.27	0.41
	< 3 d	1,276	510	0.101	−2.219	0.0108	9.27	1.54
Mentions	≥ 3 d	2,105	910	0.037	−0.006	−0.0443	0.14	0.41
	< 3 d	2,090	659	0.105	−2.049	0.0193	7.92	1.34
Counts	≥ 3 d	269	153	0.048	−0.190	0.0006	0.37	0.57
	< 3 d	363	110	0.212	−1.396	−0.0015	4.40	1.57
Eliminations	≥ 3 d	451	276	0.048	−0.306	−0.0224	0.92	0.67
	< 3 d	767	313	0.222	−2.810	0.0161	10.34	2.02

Least-squares fit of Equation (25) to $\lambda = \bar{h}$ (per day) over no-jump grid steps with both mids in $(0.02, 0.98)$ and spread at most 10¢; σ^2 is the mean squared residual per unit time. $\sigma_{\ln \bar{h}}$: standard deviation of $\Delta \ln \bar{h}/\sqrt{\Delta t}$ per square-root day. Negative κ means no mean reversion: $\bar{h}$ drifts upward towards the deadline, so θ is not a long-run level. All splits pooled.

Table 29: Economic significance: buying NO at the NO ask, held to settlement

		Estimation sample (before 1 October 2026)		Holdout (October 2026)	
Rule	Unit	Trades (cl.)	¢ (t)	Trades (cl.)	¢ (t)
Hazard model, edge ≥ 0.12	every signal	538 (143)	19.02 (5.02)	49 (13)	−5.30 (−0.83)
	first per contract	311 (143)	16.15 (4.95)	29 (13)	−8.53 (−1.26)
Price-only model, edge ≥ 0.08	every signal	444 (163)	9.83 (2.77)	27 (12)	−7.43 (−0.78)
	first per contract	344 (163)	8.20 (2.60)	20 (12)	−4.02 (−0.49)
Control, YES mid ≤ 0.50	every signal	18,949 (593)	1.16 (4.35)	1,737 (89)	1.32 (1.65)
	first per contract	3,561 (593)	1.89 (3.44)	294 (89)	0.06 (0.03)
Maker NO bid, hazard model	per fill	48 (12)	6.06 (0.68)	29 (13)	−3.17 (−0.46)
Maker NO bid, mid	per fill	32 (11)	6.56 (0.63)	14 (9)	−10.29 (−1.02)

Profit in cents (¢) per contract. One contract per signal at the NO ask (1− YES bid) with the Kalshi taker fee $0.07\, m\, P(1-P)$ rounded up to \$0.0001, held to settlement; quotes with spread at most 10¢. The hazard model is a multiplier table by type, lead and price fitted on the estimation sample; the price-only model uses price buckets only; thresholds and the control's price cap are chosen on the estimation sample. Maker rows rest a NO bid at $\lfloor 1 - p - \text{margin} \rfloor$ until the next grid point and count a fill only if a later trade prints through it (shared trade tape from 30 July 2026). t is clustered by event. In a logistic blend of the outcome on the logits of the mid and of the hazard model, the hazard coefficient is 0.006 ($t = 0.04$) in the holdout; holdout Brier scores are 0.0459 (mid) and 0.0472 (hazard model).